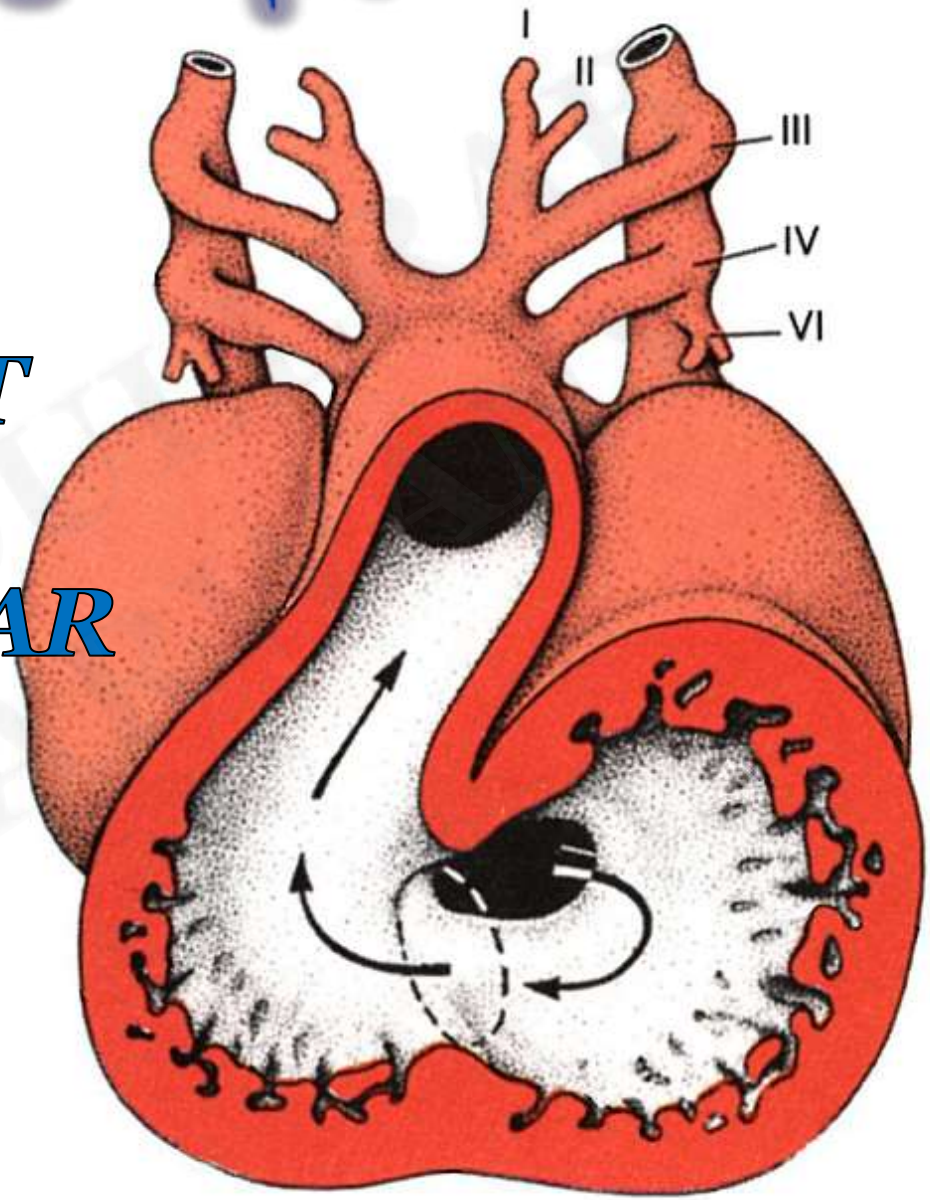
بسم الله الرحمن الرحيم

DEVELOPMENT OF CARDIOVASCULAR SYSTEM



DR. ARAB AL-MARWANY

INTRODUCTION

*The nutritional supply of embryo **changes** according to development as follows:*

- ***During the first week**, before implantation, **deutoplasm** (accumulated cytoplasmic nutrients) of oocyte supplies nutrition.*
- ***During the second week**, breakdown products of **endometrium** (due to implantation) nourishes the embryo by simple diffusion.*
- ***After the third week**, **maternal blood** nourish embryo through uteroplacental circulation.*

- *Increasing nutritional demand of growing embryo initiates development of cardiovascular system in the third week of intrauterine life.*

- *All components of cardiovascular system develop from **Mesoderm**.*
- *The components of cardiovascular system can be studied as*

1. Development of heart

2. Development of blood vessels

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Development Of The Heart

DR ABDULLRAB AL MARWANY

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Lets Remember The Result Of The Head Folding &Intra-embryonic coelom:

Intra-embryonic coelom:

Definition:

It is a horse shoe cavity that is formed in the lateral plate mesoderm.

- *Its caudal end **communicate** with the extra-embryonic coelom at the margin of embryonic disc.*

- *It is **divided into**:*

***a) Pericardial cavity** in median cranial part of the coelom.*

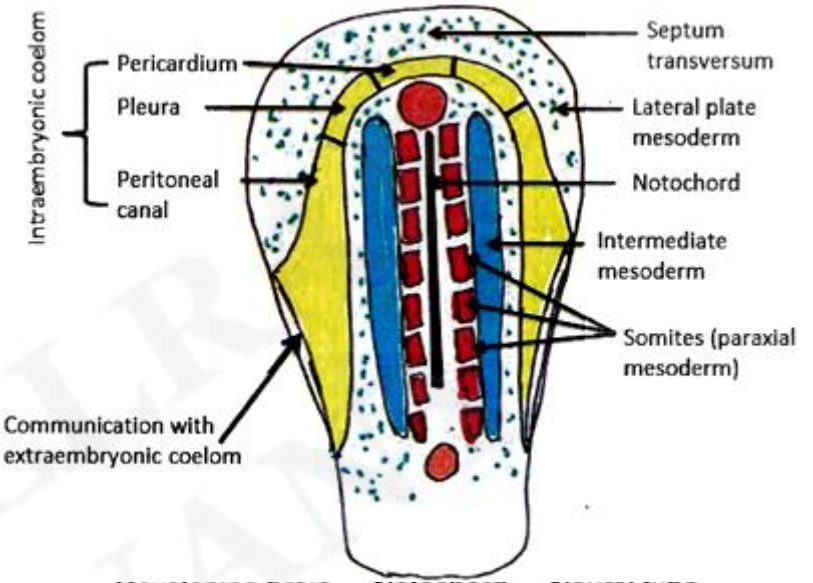
***b) Peritoneal canals** in the lateral part of the coelom.*

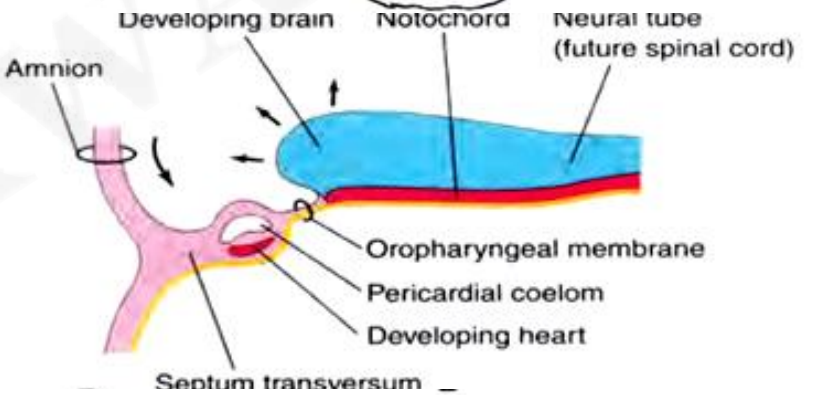
***c) Pleural cavities** between pericardium and peritoneal canals.*

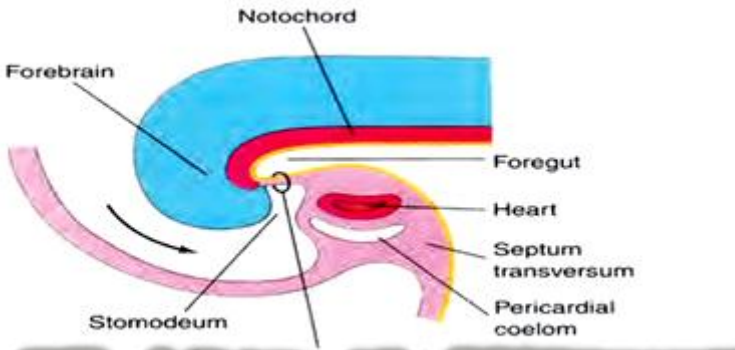
N.B.

Cardiogenic area is the splanchnic mesoderm related to pericardium.

*Septum transversum is a mass of mesoderm **cranial** to pericardium (**before folding**) and **caudal** to pericardium (**after folding**).*







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I- Establishment Of The Cardiogenic Field:

Sources Of The Heart Development

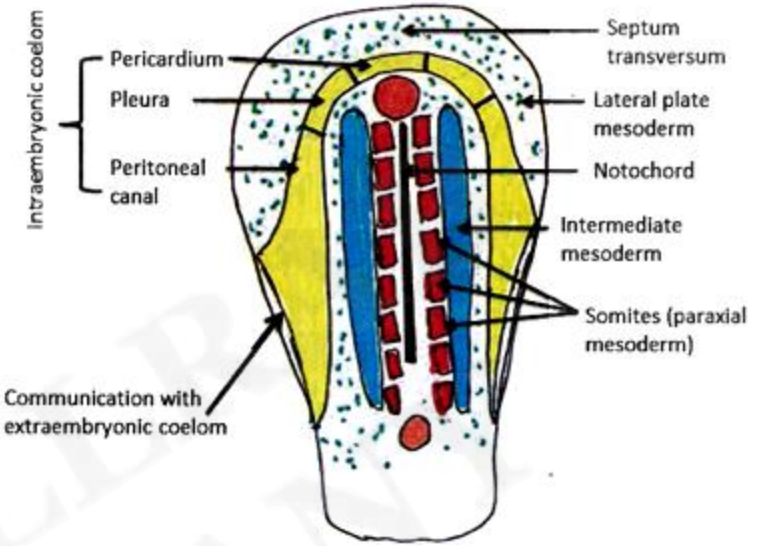
1- Pericardial sac

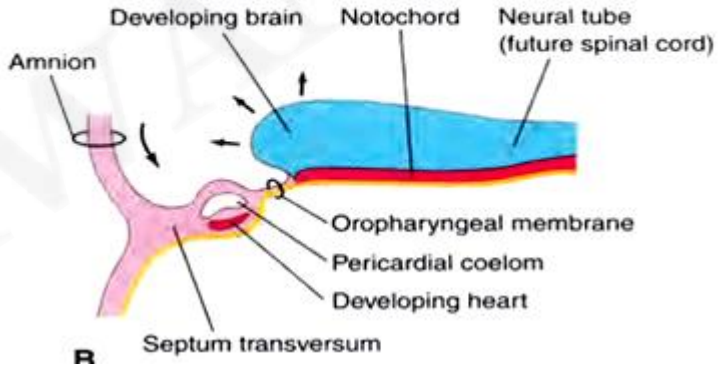
2- Endocardial heart tube:

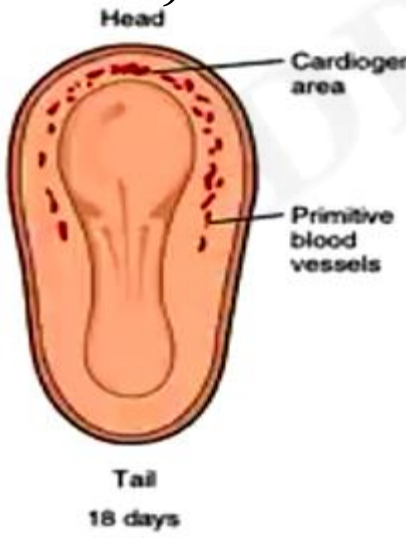
3- Myoepicardial mantle :

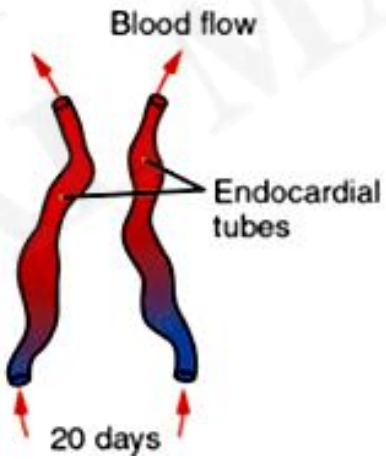
1- Pericardial sac : *It is the transverse (cranial) part of the intraembryonic coelom.*

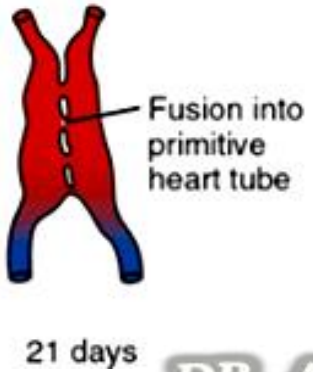
2- Endocardial heart tube: (*Angiogenic cells appear at the area of cardiogenic plate to form Rt & Lt endocardial heart tubes. Both fuse together forming a single tube).*

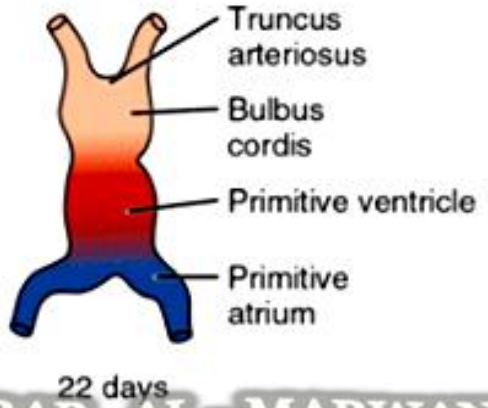












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The Progenitor heart cells:

***Lie** **in the epiblast**, immediately adjacent to the cranial end of the primitive streak.*

- ***During days 16 to 18:***

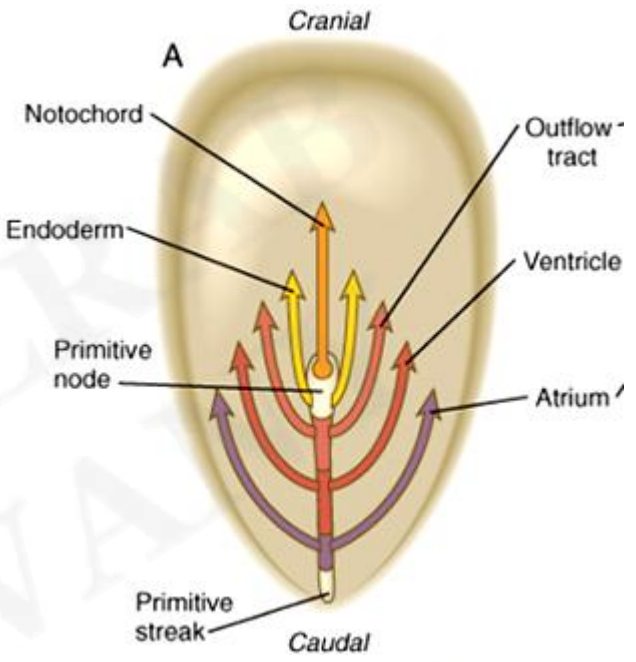
➢ *The Progenitor heart cells **migrate** through the streak and **into** the **splanchnic layer** of lateral plate **mesoderm** where they **form** a horseshoe-shaped cluster of cells called the **Primary Heart Field (PHF)** **cranial to the neural** folds.*

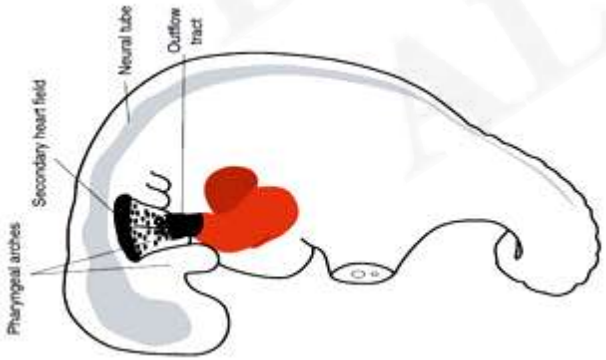
- *As the progenitor heart cells migrate and form the PHF, they are **specified** on both sides from lateral to medial to become the **atria**, **left ventricle**, and **most of the right ventricle** .*

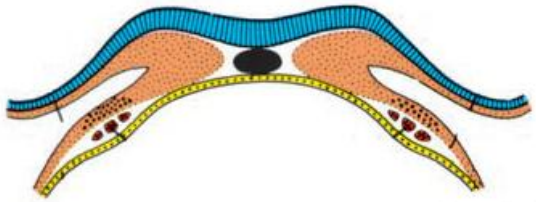
- ***During days 20 to 21:***

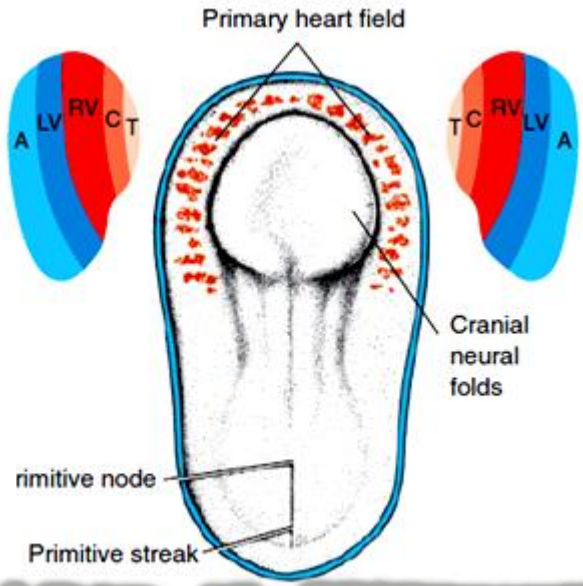
➢ *The **Secondary Heart Field (SHF)** **of** cells appears & resides in splanchnic mesoderm ventral to the posterior pharynx.*

- *It is responsible for development of the remainder of the heart, including part of the **right ventricle** and lengthening the **outflow tract** (conus cordis and truncus arteriosus)**.***



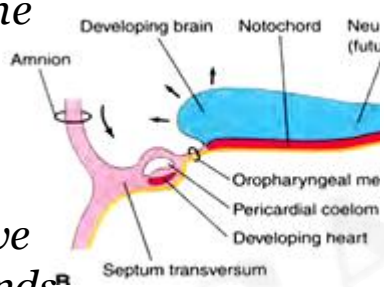


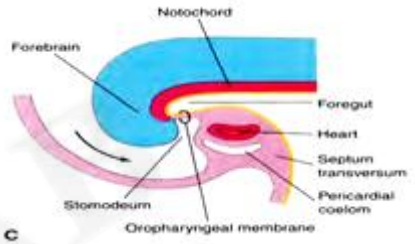




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- *On formation of head fold, primary heart field come to lie on dorsal side of pericardial sac.*



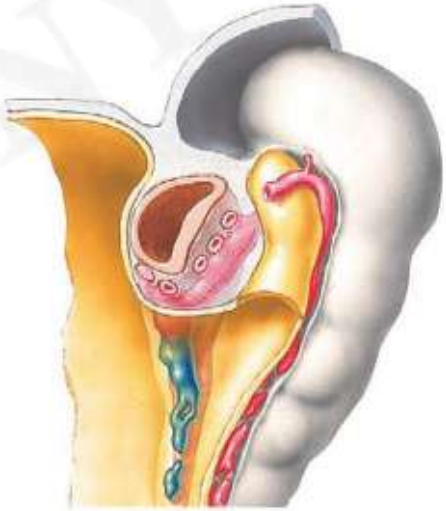


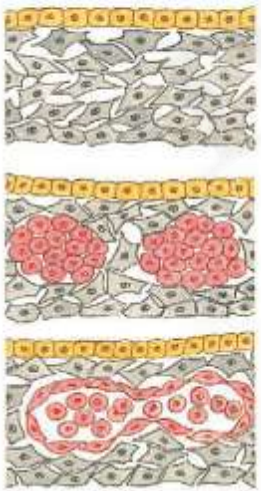
- *By the process of **vasculogenesis** the Progenitor **heart** cells are induced by the endoderm of primitive pharynx to form cardiac myoblasts and blood islands that will form blood cells and vessels .*







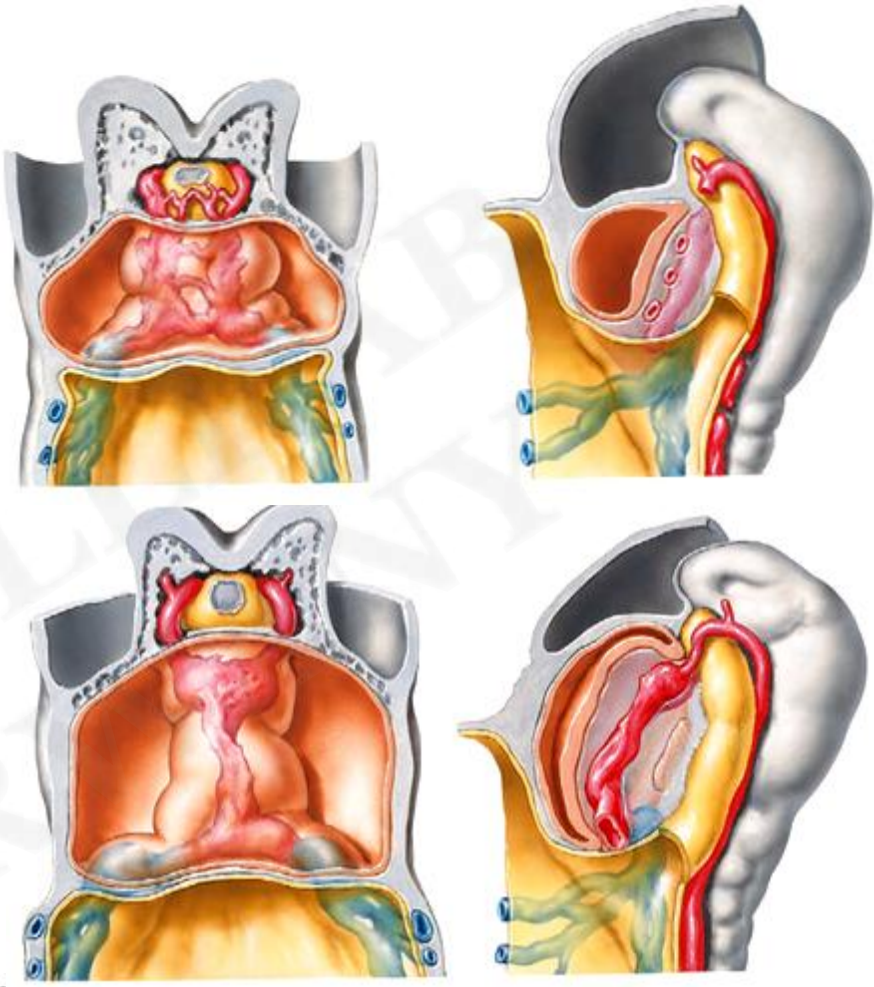




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- *With time, the islands unite and form Small vessels that join to form two (right and left) endothelial heart tubes that give rise to the endocardium surrounded by myoblasts.*

- *The two heart tubes fuse and form a single heart tube. Ends of the heart tube remain bifurcated. Its cranial end is called arterial end **outflow**, whereas caudal end is called venous end **inflow.***



- ***The outflow to the heart tube:***
- *A paired dorsal aortae form simultaneously with the endocardial heart tubes.*
- *They are connected cranially with the endocardial tubes.*

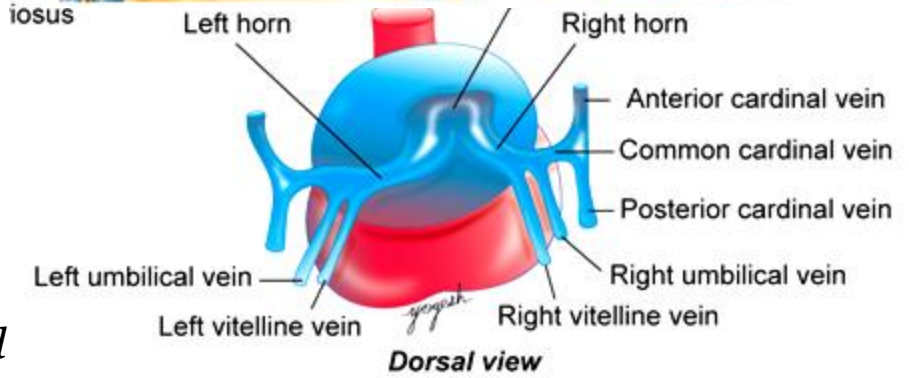
- ***The inflow to the heart tube:***

Is supplied by three pairs of vessels:

a) *A pair of **common cardinal veins** drain blood from the body of the embryo.*

b) *A pair of **vitelline veins** drains the yolk sac.*

c) *A pair of **umbilical veins** carries oxygenated blood from the placenta.*



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3- Myoepicardial mantle :

After invagination of the pericardial sac by the heart tube ***the Splanchnopleuric mesoderm (that lies*** *between them) will* form the *myoepicardial mantle*.

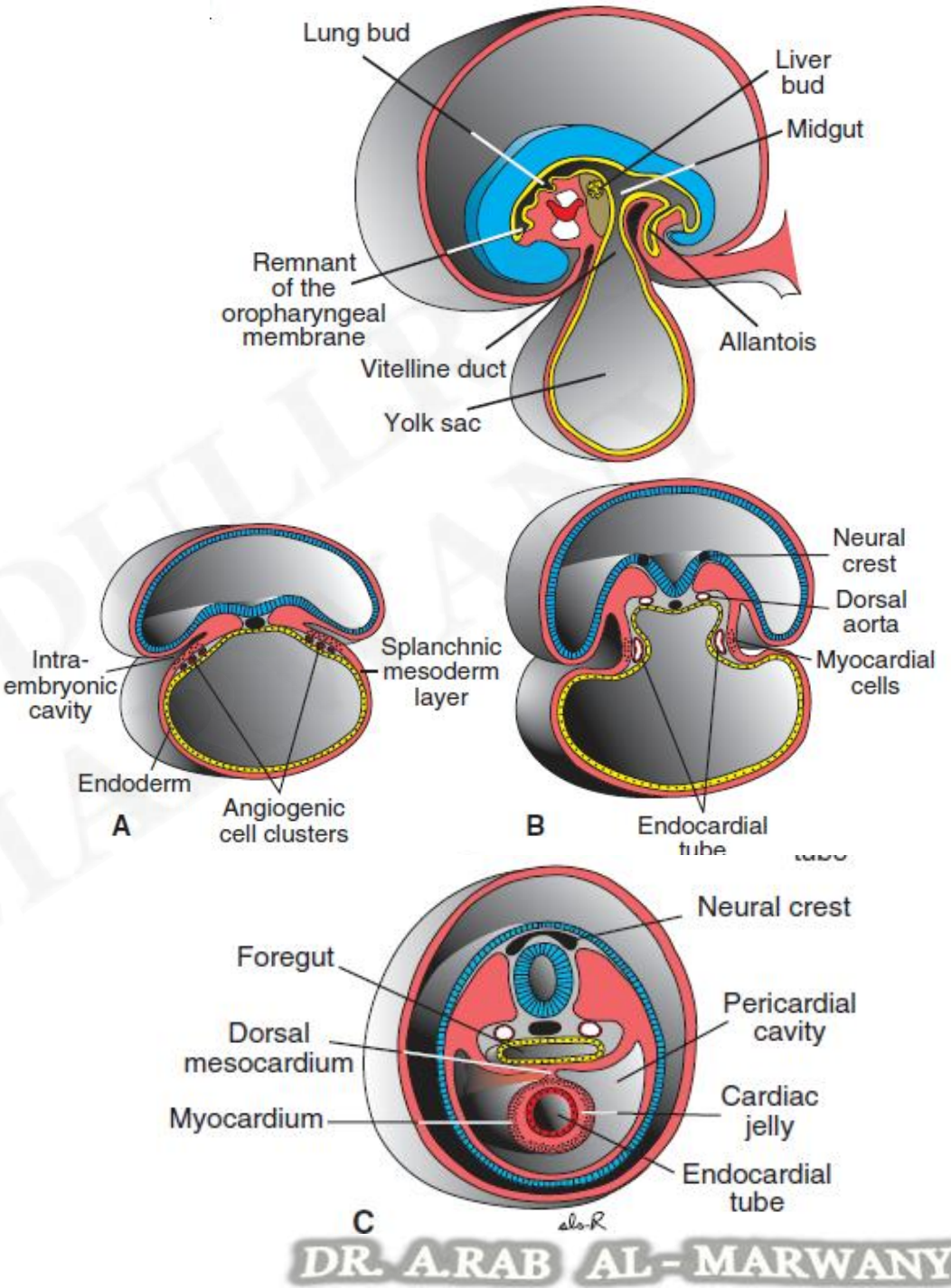
The myoepicardial mantle condense to form

– ***Myocardium*** *(cardiac muscles) [cardiac muscles develop from Splanchnopleuric mesoderm.*

– ***Epicardium*** *(**visceral** layer of* ***pericardium***) .

*Somatopleuric mesoderm that surrounds pericardial cavity form **parietal** layer of* ***pericardium***.

- *Primitive heart* ***starts beating*** *on the **22nd** day.*
- *Blood begins to* ***circulate*** *within the embryo by **24th** day*



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II- Changes Of The Heart Tube

Heart tube undergoes growth and folding to acquire external features of the adult heart as follows:

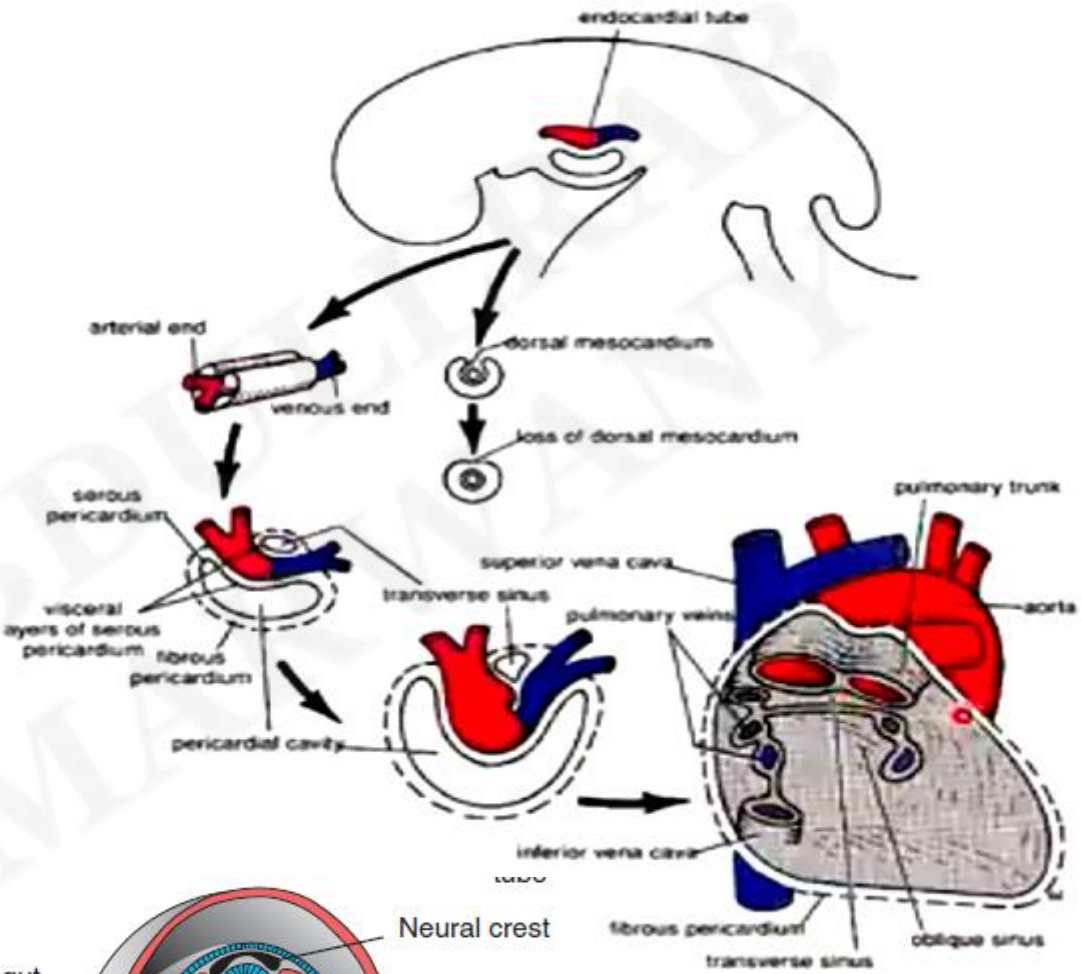
1. Invasion Of The Pericardiac Sac :

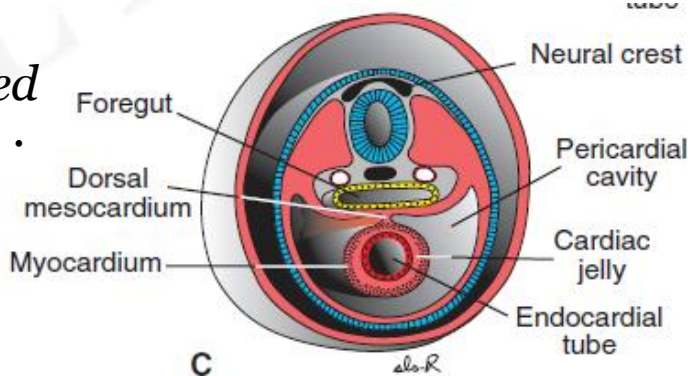
- *It occurs from the dorsal aspect of the pericardial sac .*
- ***Formation of transverse sinus:***

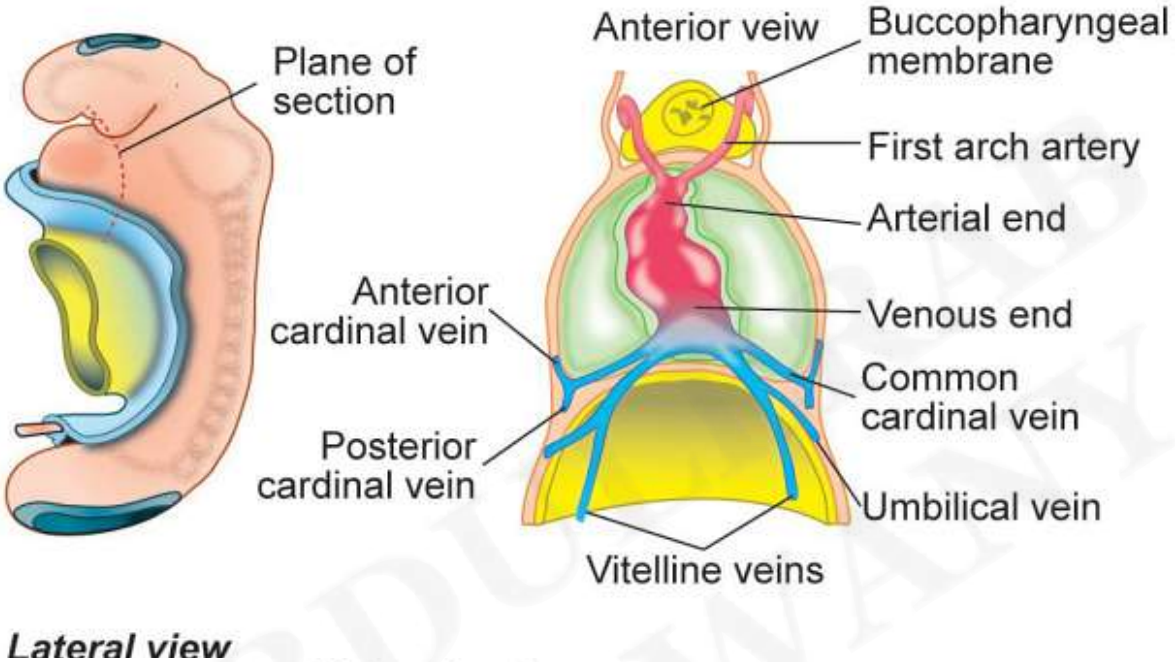
➢ ***Initially**, heart tube is placed longitudinally in pericardial cavity.*

➢ *Heart tube is <u>suspended</u> from dorsal wall of pericardial cavity by a fold of pericardium called **dorsal mesocardium** .*

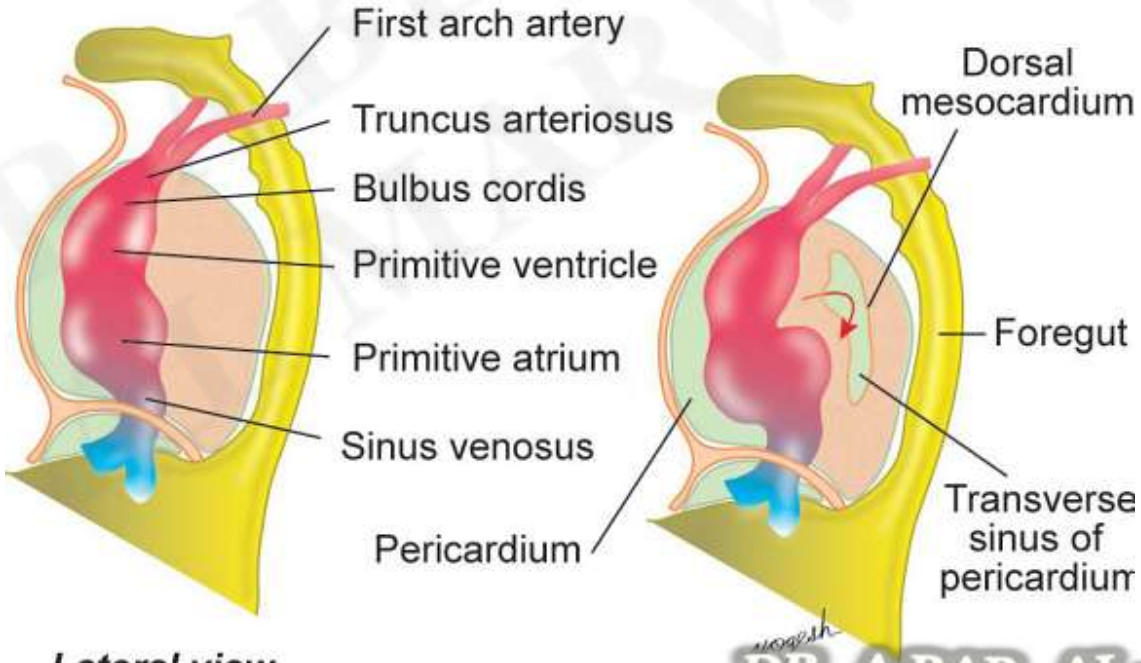
➢ *The mesocardium **disappear** to **form** a gap, that later called **transverse sinus** .*







Lateral view



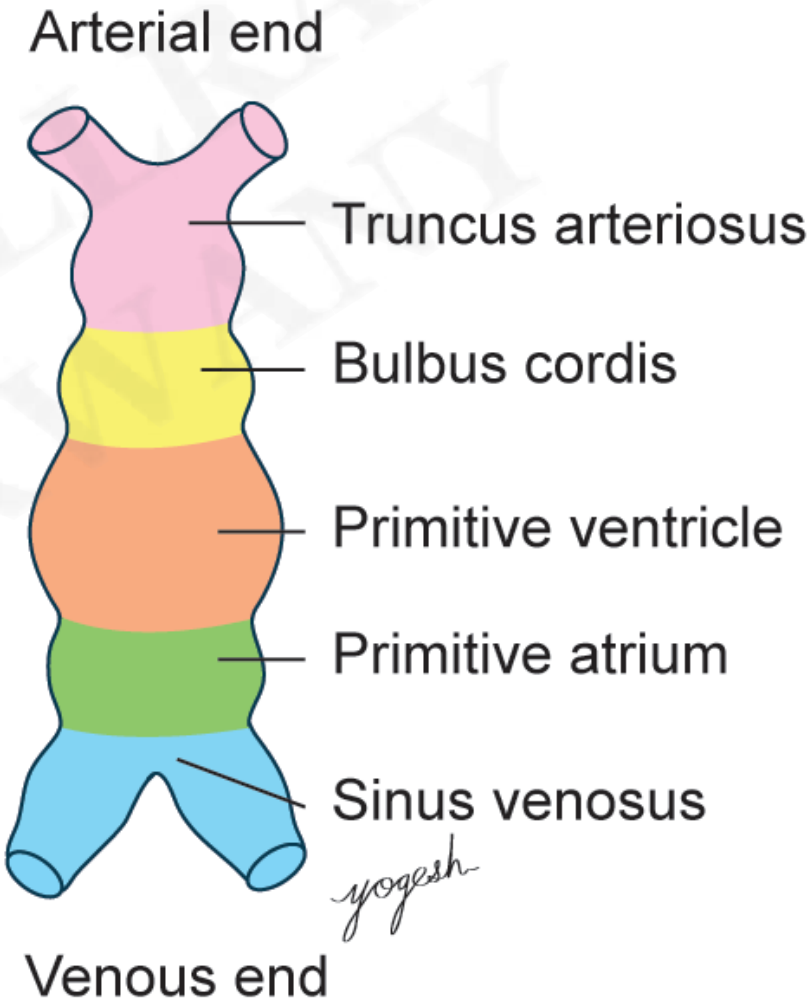
Lateral view

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2- A Series Of Constrictions Appear In The 1ry Heart Tube:

- *They subdivide the heart tube into the **five** heart dilatations (from cranial to caudal:*

- ***Truncus arteriosus.***
- ***Bulbus cordis**, is separated from the ventricle by the bulboventricular sulcus.*
- ***Primitive ventricle** is separated from the atrium by the atrioventricular canal.*
- ***Primitive atrium.***
- ***Sinus venosus**: consists of a central part receiving left and right horns.*



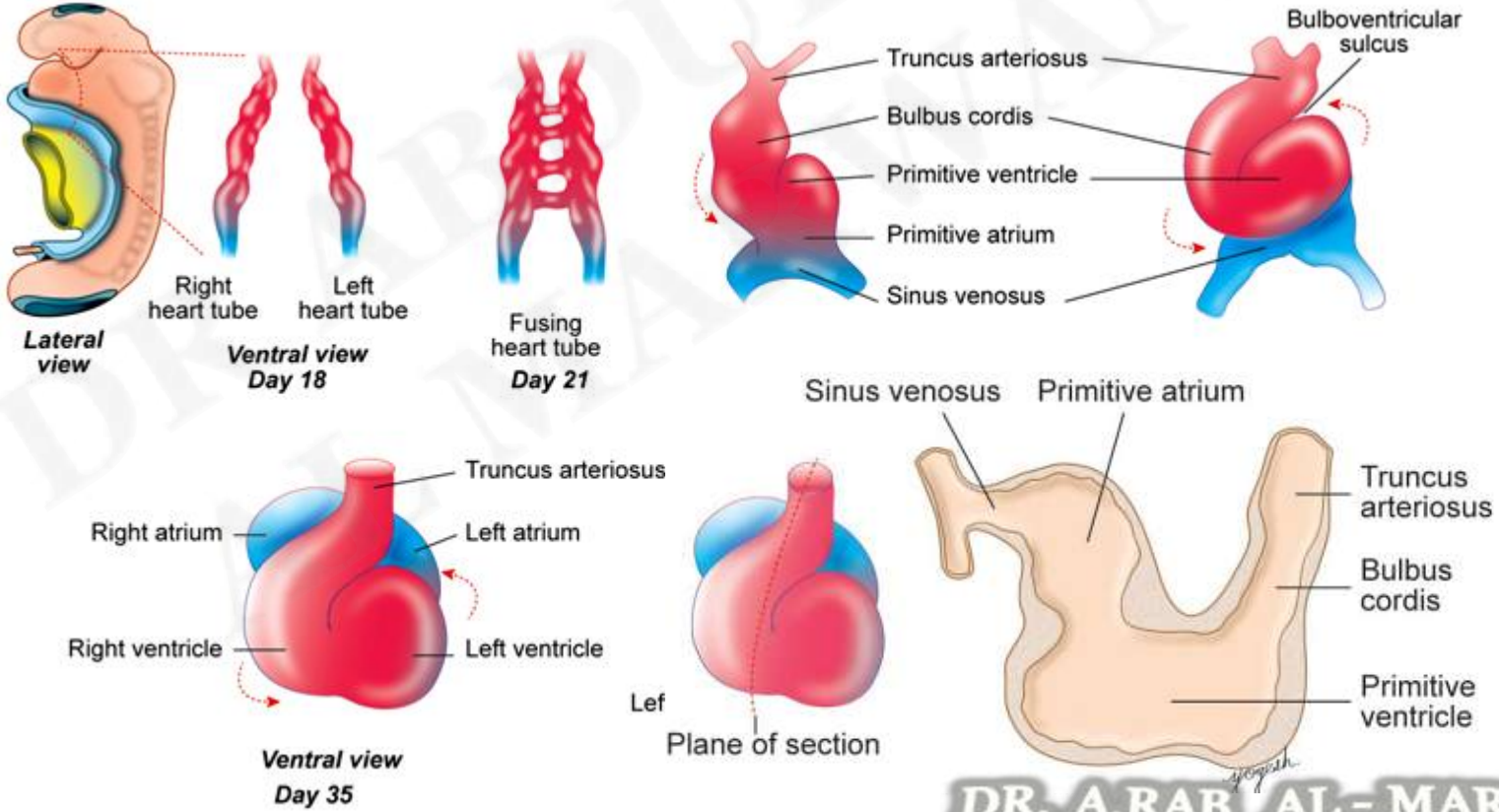
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3- Formation Of The Cardiac Loop:

*The heart tube begins to **elongate** and simultaneously to **loop** and **fold**.*

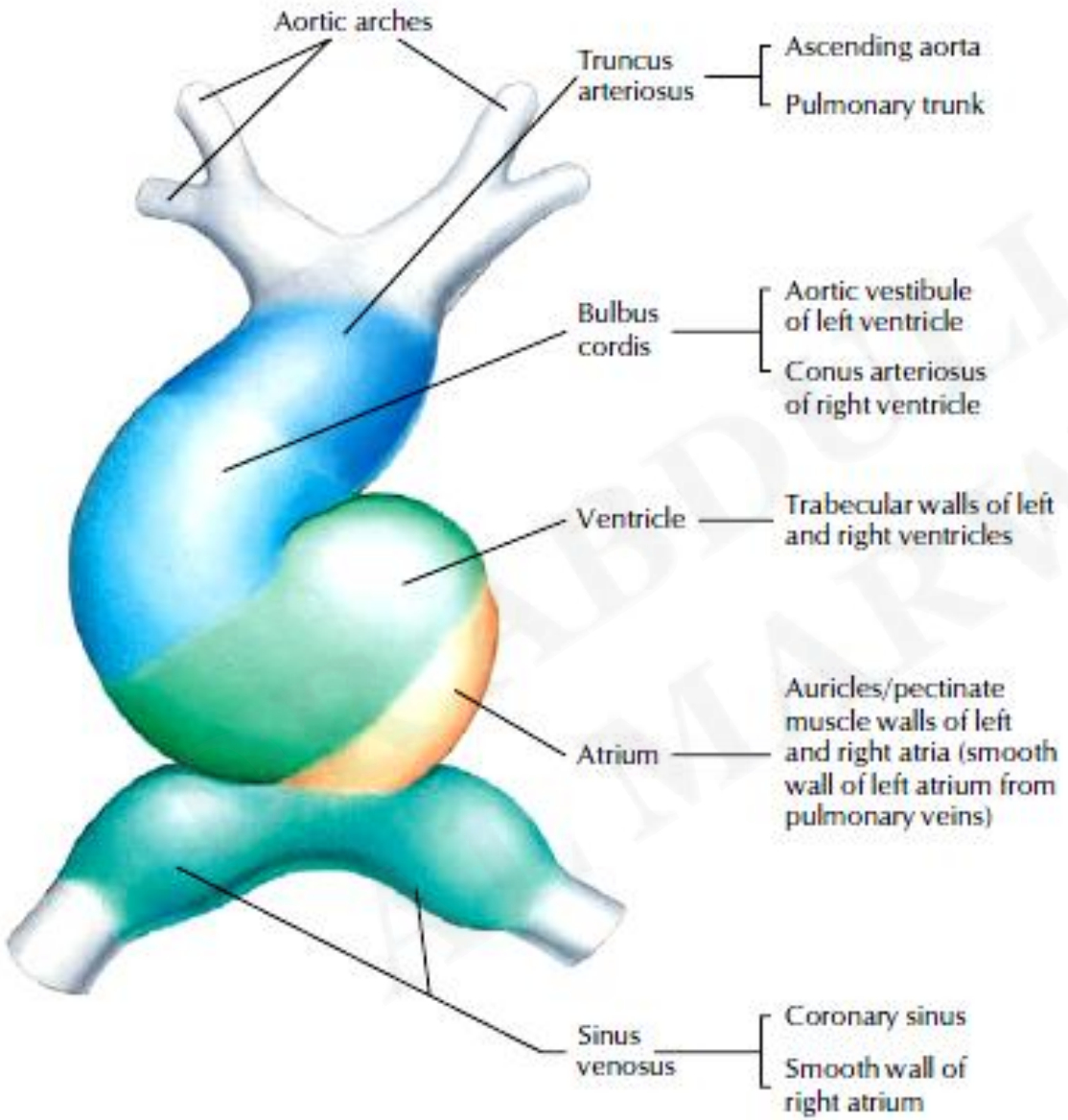
- *The **bulbus cordis** is displaced inferiorly, ventrally, and to the right & the **primitive ventricle** is displaced to the left ,forming the bulboventricular loop (U-shaped).*
- *Bulbus cordis and primitive ventricle are separated by bulboventricular sulcus that later disappears and bulbus cordis and ventricle fuse to form a single chamber .*

** The **primitive atrium** is displaced posteriorly and superiorly. S-shaped cardiac loop is formed. On expansion, primitive atrium project forward on either side of truncus arteriosus as auricles.*

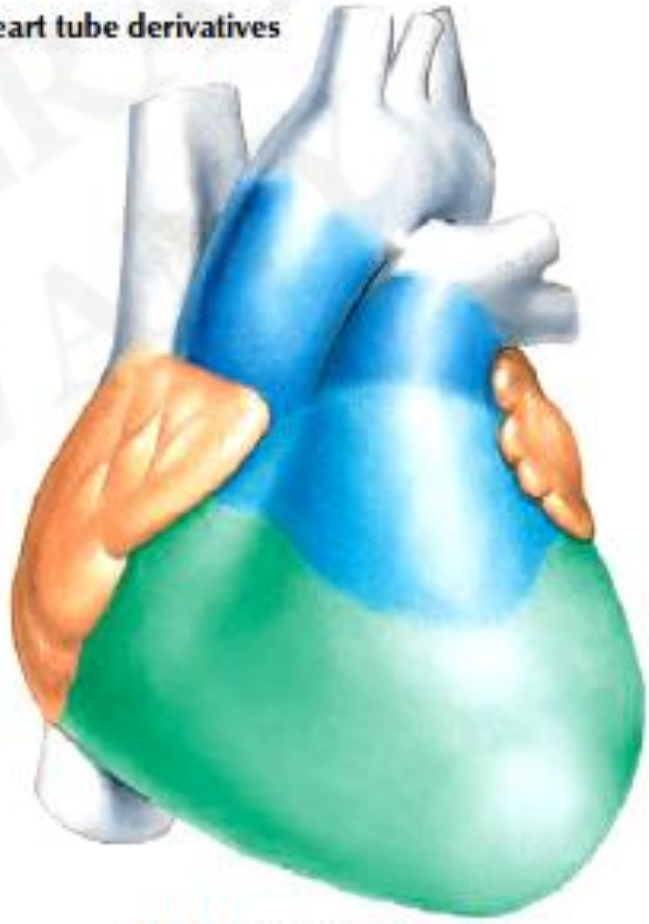


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Heart tube primordia



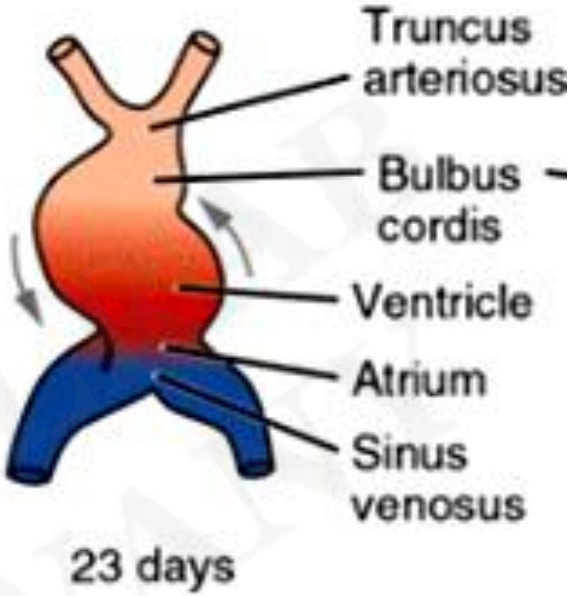
Heart tube derivatives



Adult heart, anterior view

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Sinus Venosus



➢ ***Definition:***

- *It is the most caudal part of the heart tube.*
- *It receives all veins of the body.*

➢ ***Structure :***

At first **:** *It is formed of a small median part and two lateral horns* ***(Rt and Lt).***

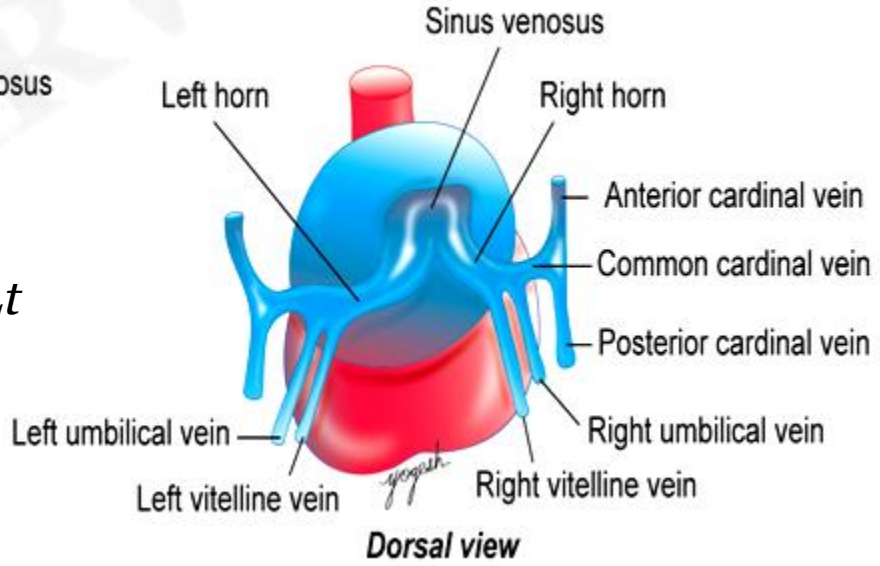
Each horn receives 3 veins :

a) Vitelline vein: *from yolk sac.*

b) Umbilical vein: *from the placenta.*

c) Common cardinal vein: *from the body of the embryo itself.*

iosus



➢ *It opens into the primitive atrium by a sino-atrial orifice guarded by Rt and Lt valves.*

Sinoatrial orifice: *It is a communication between sinus venosus and primitive atrium.*

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Fate of the sinus venosus:

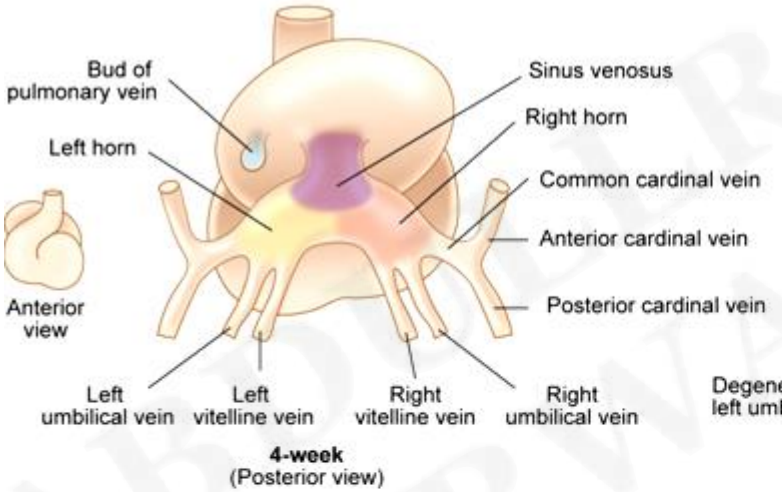
1-The left sinus horn: *diminishes in size due to:*

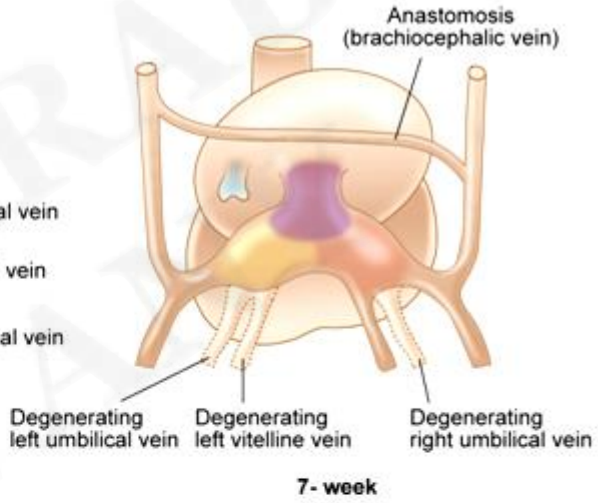
a- Lt vitelline and Lt umbilical veins lose their connection with the sinus venosus.

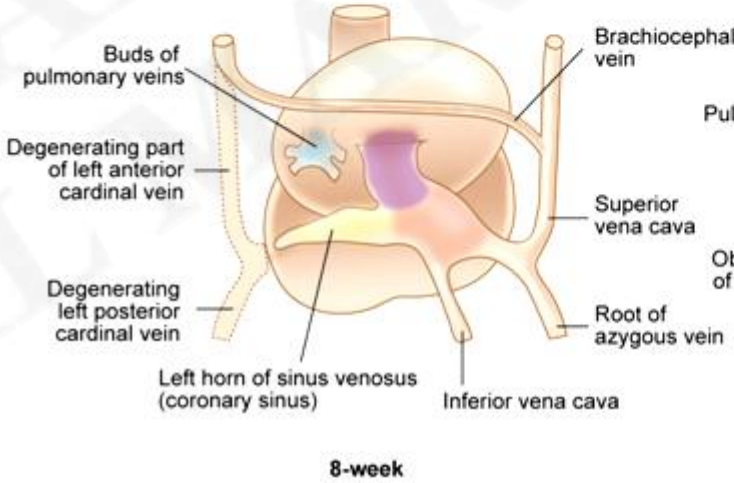
b- Shift of venous blood to the Rt side due to development of transverse anastomoses between the 2 anterior cardinal veins and 2 posterior cardinal veins.

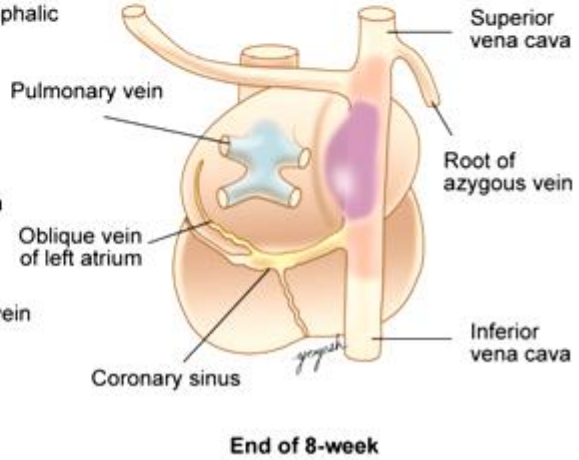
The left sinus horn becomes smaIIer and forms coronary sinus.

The left common cardinal vein forms the oblique vein of Lt atrium.





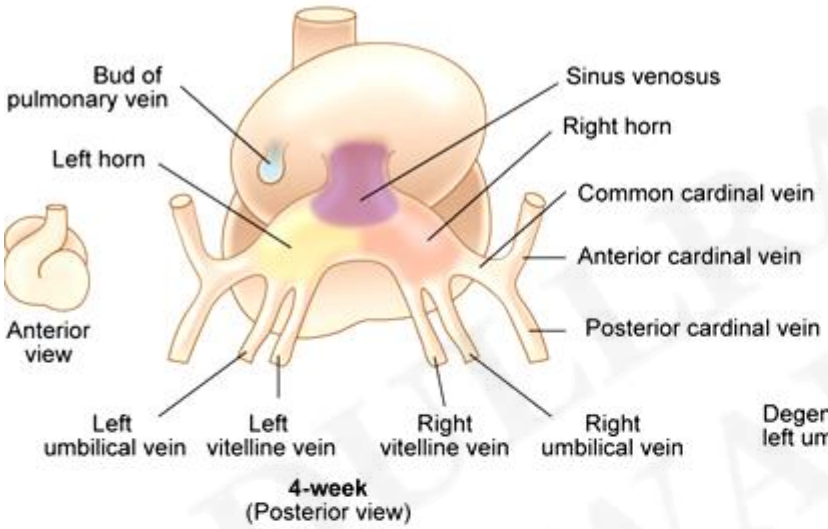


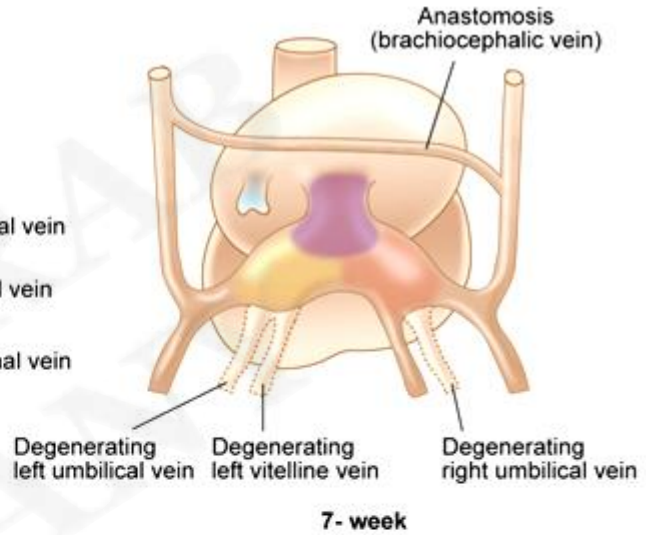


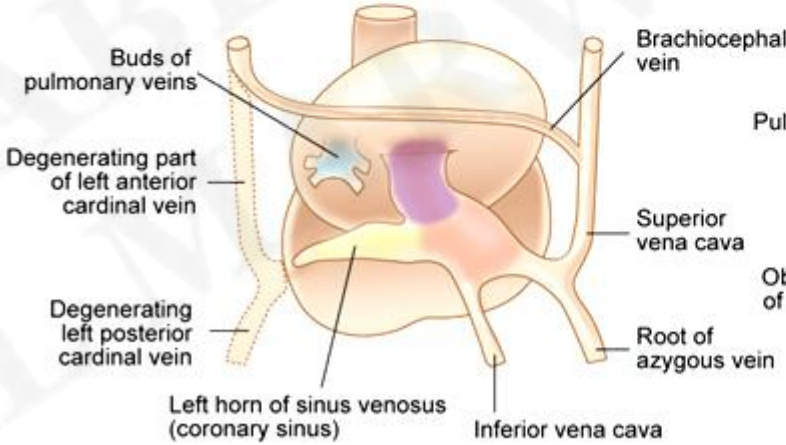
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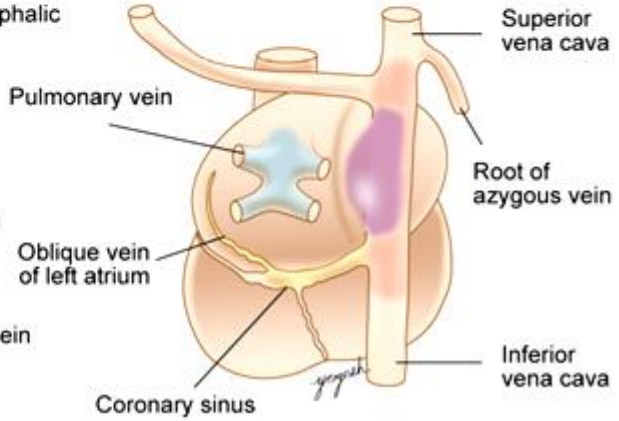
2- The right sinus horn and median part:

- *become absorbed into the Rt. side of the primitive common atrium and will form the sinus venarum (the posterior part of Rt atrium).*
- *<u>The right common cardinal</u> vein forms the lower part of SV.C.*
- *<u>Right vitelline vein</u> → terminal part of inferior vena cava.*
- *<u>Right umbilical vein</u> and left vitelline vein - →obliterated by 5th week.*









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3- Rt & Lt valves of the sino-atrial orifice:

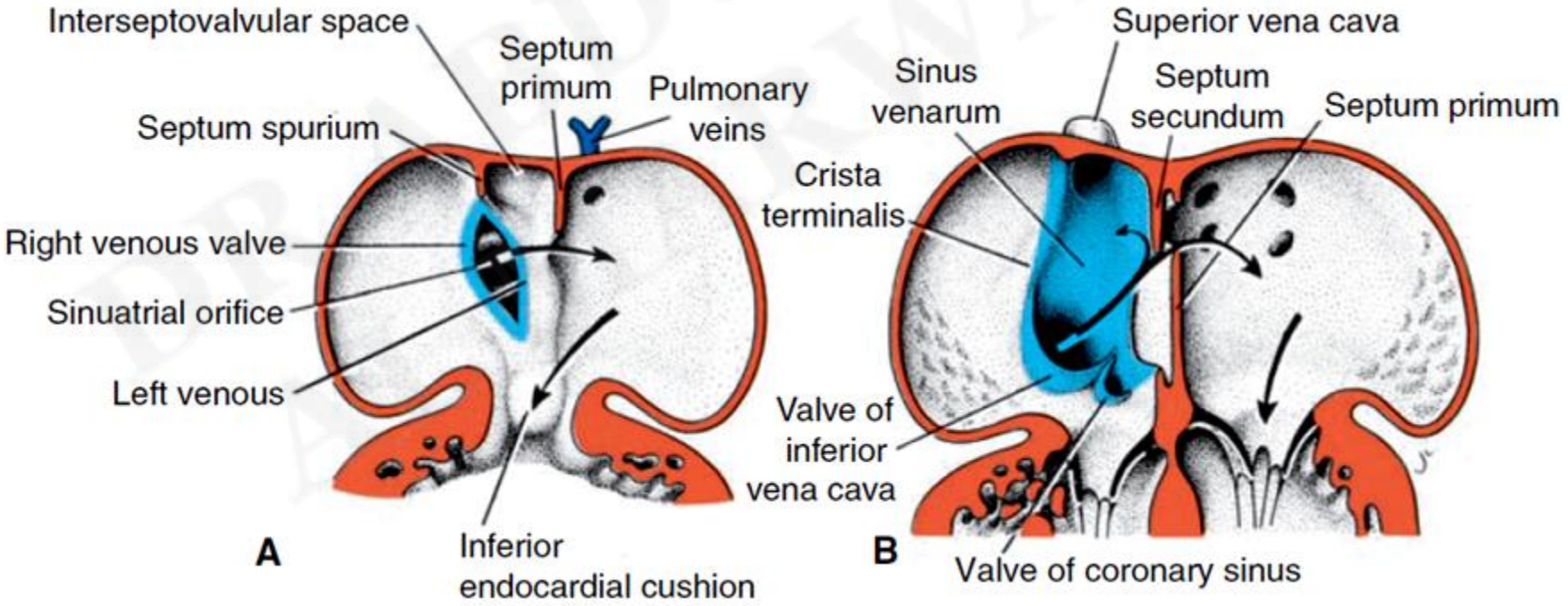
- *These are 2 folds which appear at the sino-atrial orifice.*
- *They fuse cranially to form the Septum spurium which forms the upper part of the crista terminalis.*

- *The sino-atrial orifice has 2 valves:*

* ***Lt valve*** *fuses with inter atrial septum*

* ***Rt valve****: Its upper part --→ lower part of the crista terminalis.*

Its lower part -→ valves of IVC & coronary sinus



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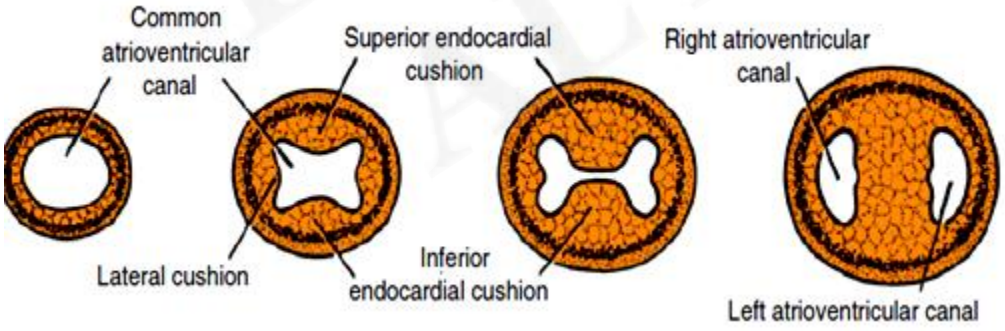
Development of the atri

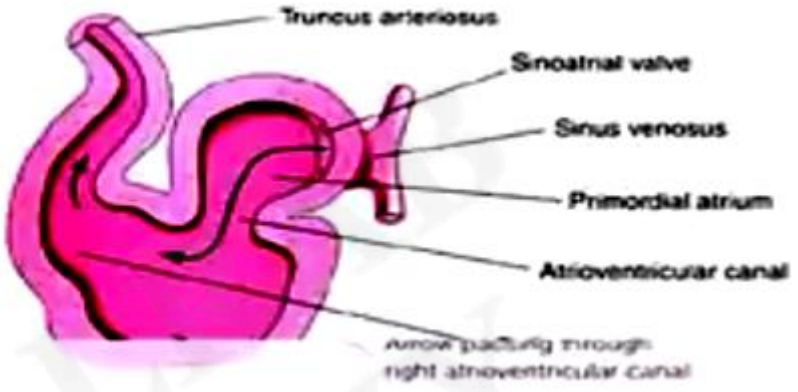
1- Development of the interatrial septum

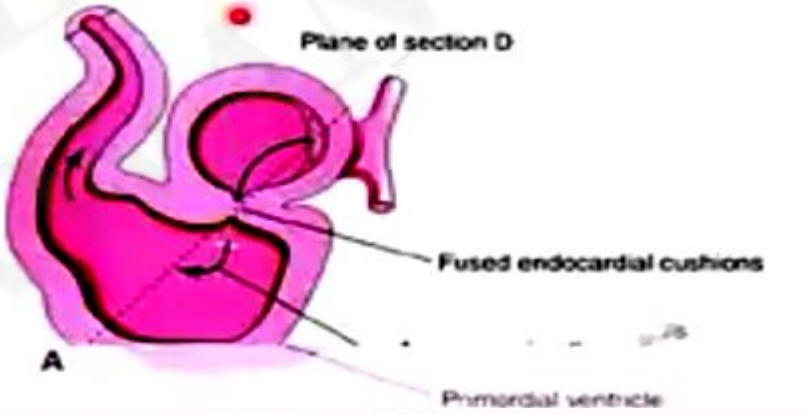
The interatrial septum develops from 3 sources:

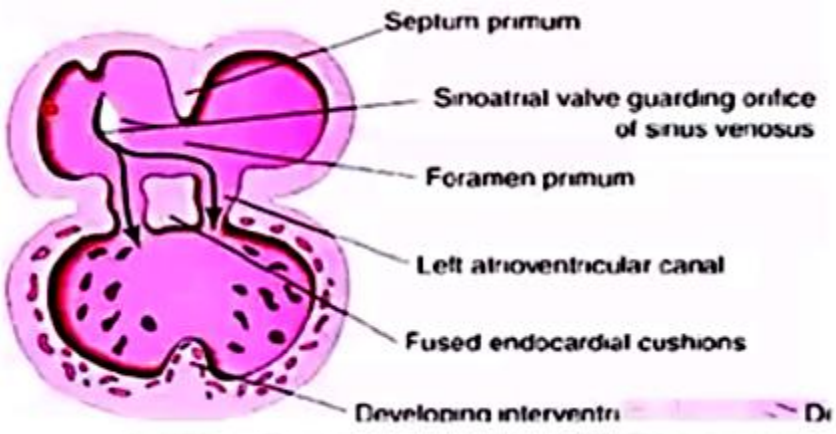
a) Septum intermedium in the atrioventricular (AV) canal

- *Ventral & dorsal endocardial cushions appear in walls of A-V canal.*
- *The 2 cushions approach each other and fuse forming septum intermedium.*
- *It divides the A-V canal into 2 halves (Rt & Lt).*

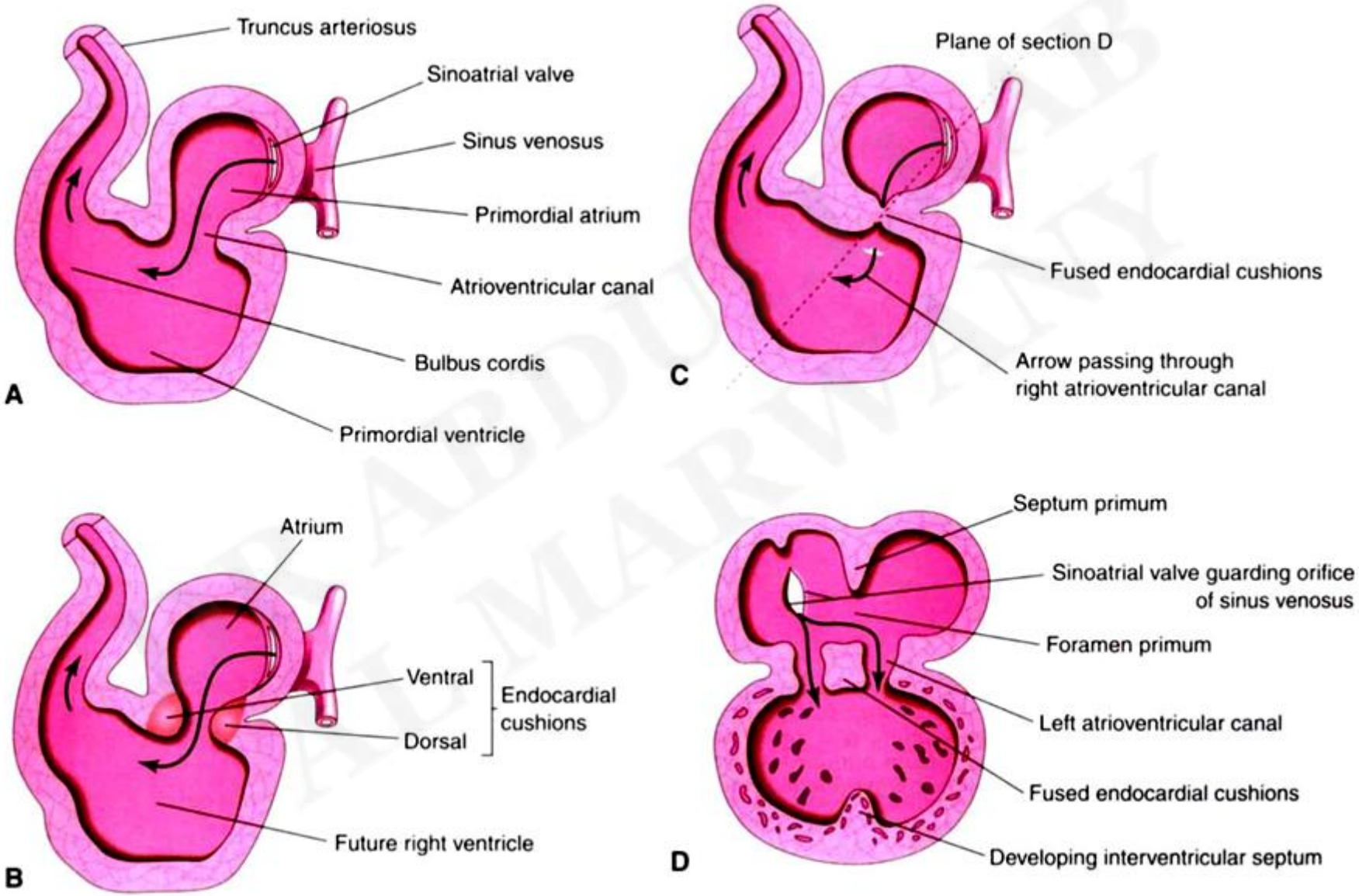








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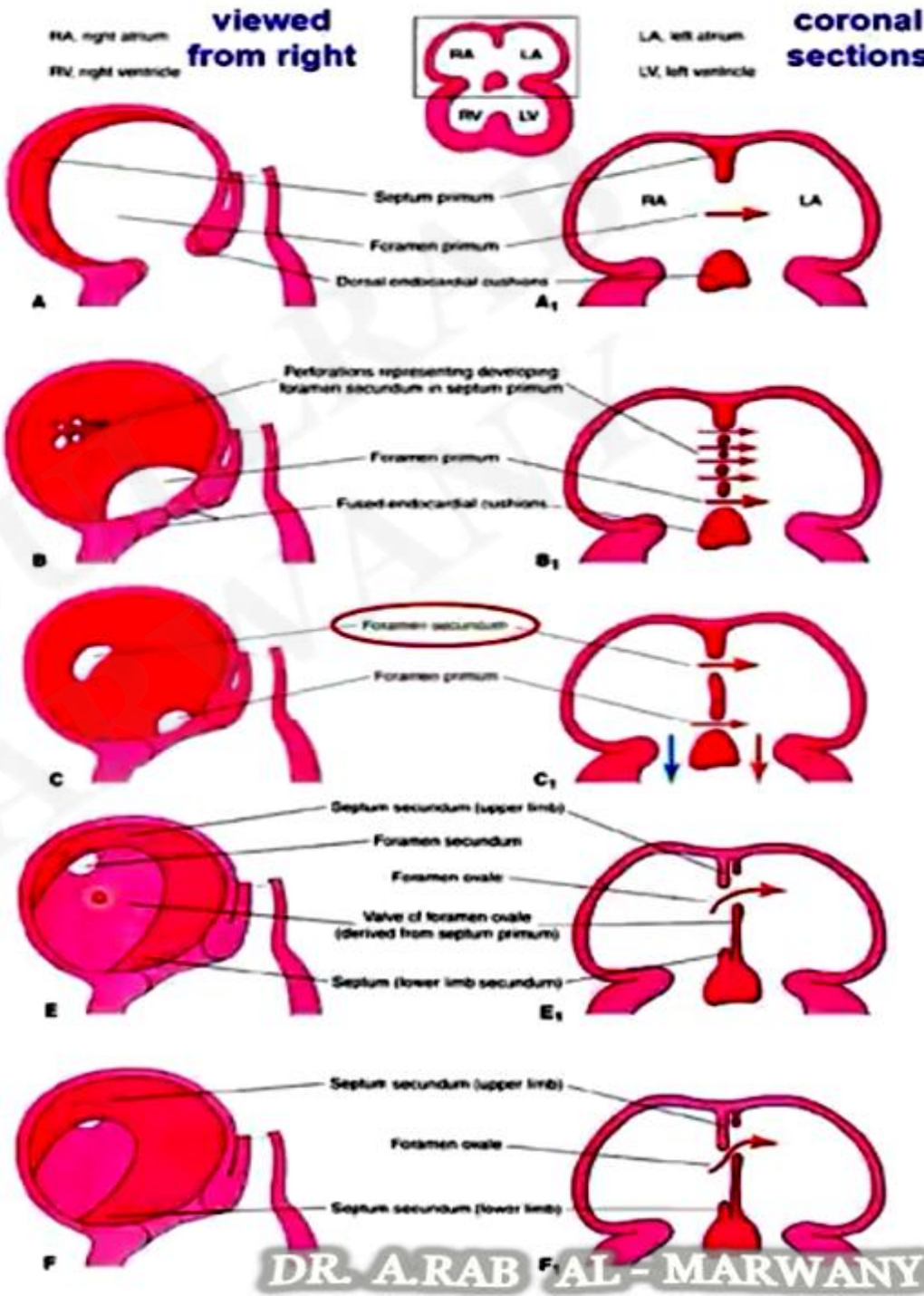
b) Septum primum

Shape: *Sickle-shaped having lowered crescentic border ending in ventral and dorsal horns.*

Origin and course: *it arises from roof of the common atrium and descends in direction of endocardial cushions of the A-V canal.*

Fate:

- *The septum primum divides the cavity of the primitive atrium incompletely into Rt. and Lt. halves.*
- *The lower border of the septum is still separated from the endocardial cushions by a gab is called the* ***ostium primum****.*
- *As development proceeds,* ***ostium primum becomes obliterated****.*
- *A second foramen called* ***ostium secundum*** *appears as a result of breaking down of the cephalic part of the septum primum.*



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c) Septum Secundum

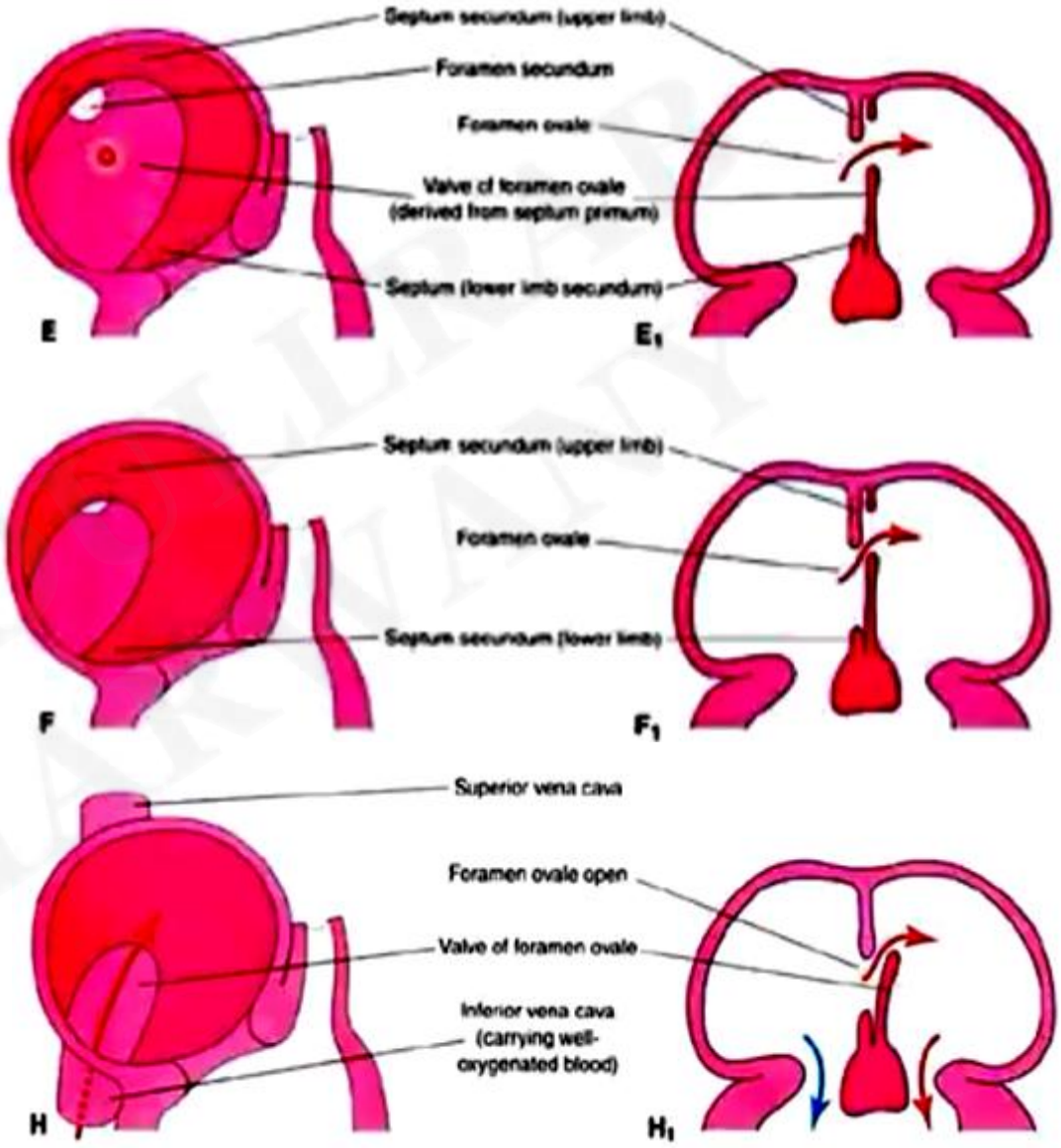
Shape: *Sickle-shaped having a lower crescentic edge with 2 horns (ventral and dorsal).*

Origin and course:

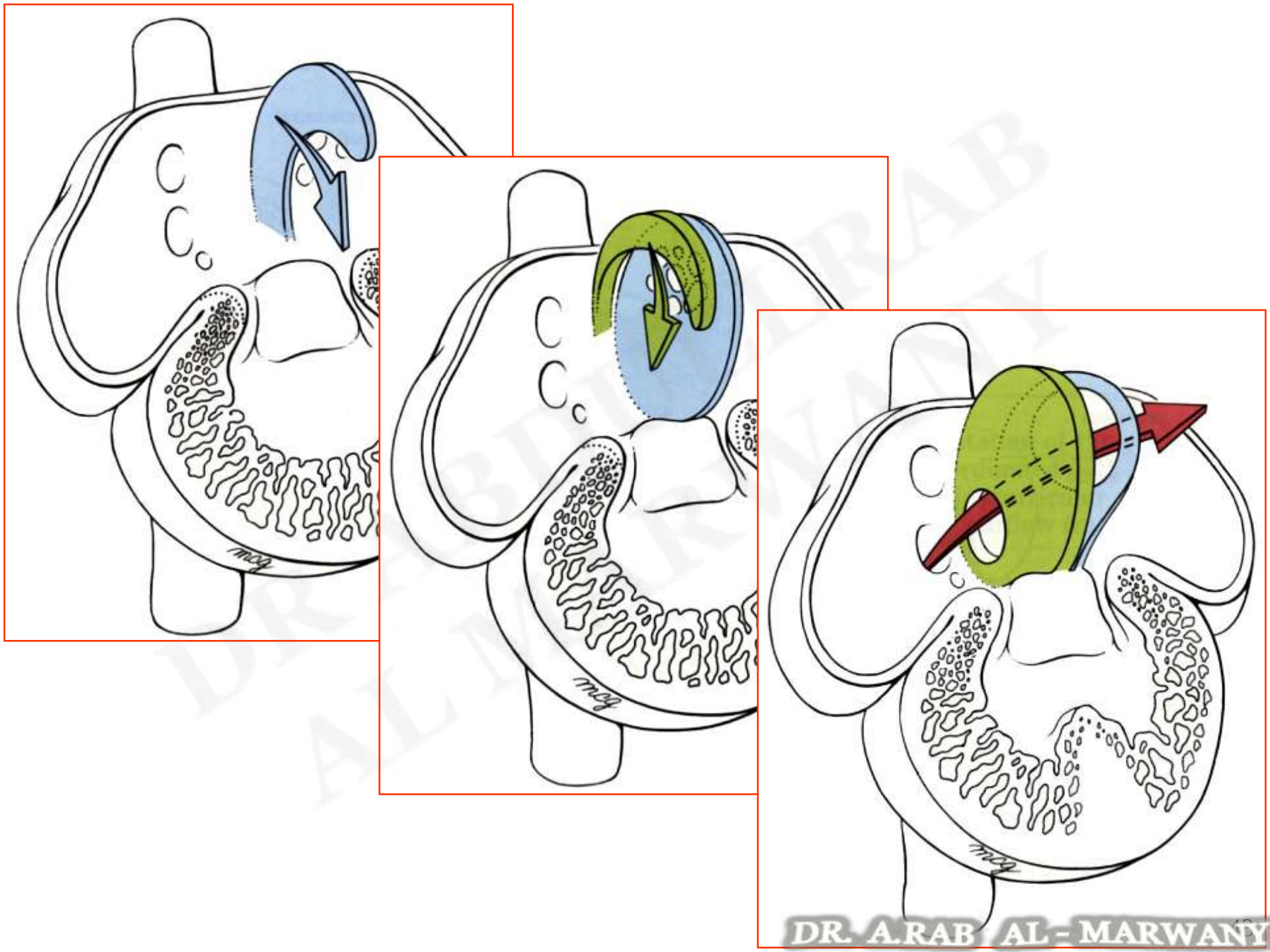
- *It arises from the roof of the common atrium (on the Rt. side of the septum primum).*
- *It descends caudally in the direction of the septum intermedium until its 2 horns fuse with it.*

Fate:

- *The septum secundum overlaps the ostium secundum. Its caudal edge is still separated from the upper edge of the septum primum by a gab is called the foramen ovale.*
- *The foramen ovale closes at birth due to firm apposition and fusion between septum primum & secundum to form the interatrial septum.*



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Embryological remnants of the two fetal septa in the adult heart

Fossa ovalis: *represents the lower part of the septum primum.*

Annulus ovale: *represents free caudal edge of the septum secundum.*

1- Absorption of the A-V canals in to the atria

Each of the Rt. and Lt. A-V canals becomes absorbed into the corresponding atrium.

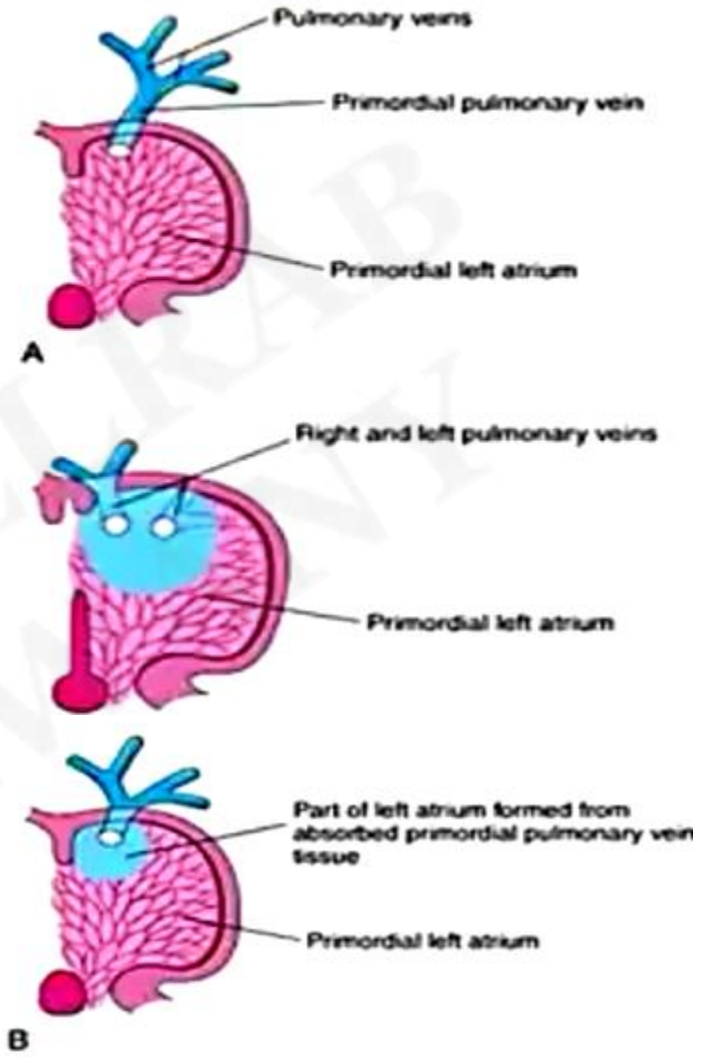
2-Absorption of the pulmonary vein into the left atrium:

- At first, a single pulmonary vein opens into the dorsal wall of the Lt. atrium.

- This single vein is formed by the union of Rt and Lt vein, each of which is formed by 2 veins emerging from the lung.

- Later on, the stem of the common pulmonary vein and its 2 tributaries become absorbed into the wall of the Lt. atrium.

- Thus the 4 pulmonary veins open separately into the Lt. atrium.



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Anomalies of the interatrial septum

1·Premature closure of foramen ovale:

Character: *premature closure of foramen ovale in the intrauterine life.*

Results: *hypertrophy of Rt atrium and Rt ventricle with underdevelopment of Lt ventricle. It leads to the death shortly after birth.*

2- Probe patent foramen ovale:

Normally, the foramen ovale closes completely at birth.

In 20 - 25 % of people, a probe can be passed through the foramen ovale.

3- Ostium secundum defect:

Character: *there is a large opening between the 2 atria.*

Cause: *it is due to either:*

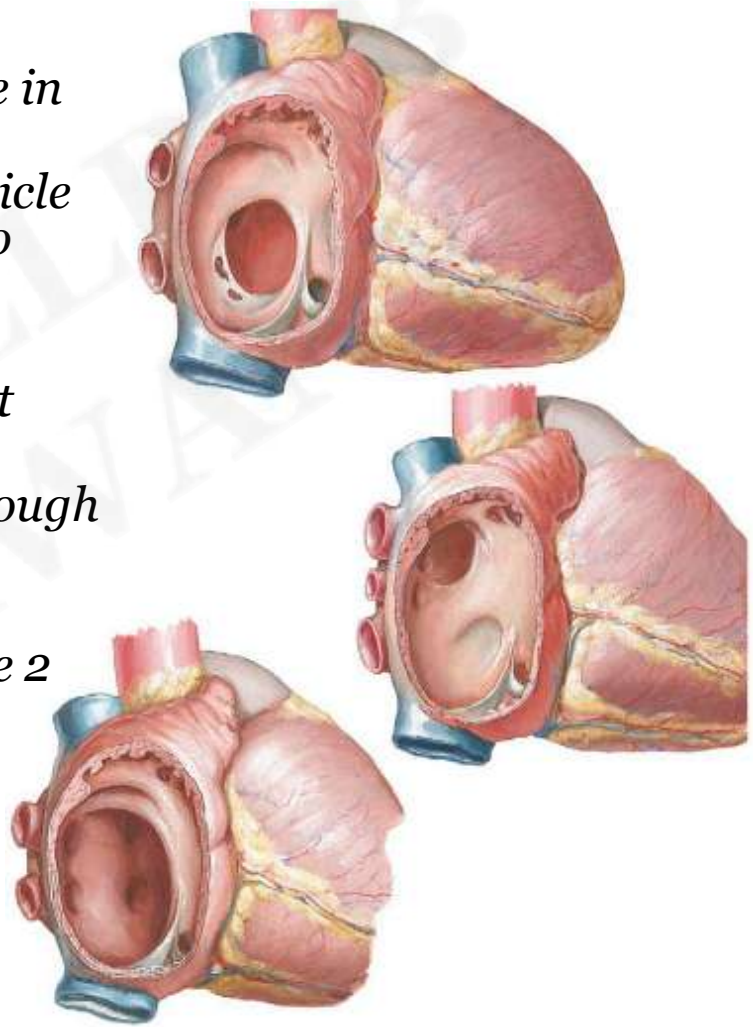
-Excessive resorption of septum primum.

-Improper formation of septum secundum.

4- Complete failure of formation of the interatrial septum:

The heart is formed of 3 chambers (common atrium & 2 ventricles).

Cause: *it is due to failure of septation of the primitive atrium.*



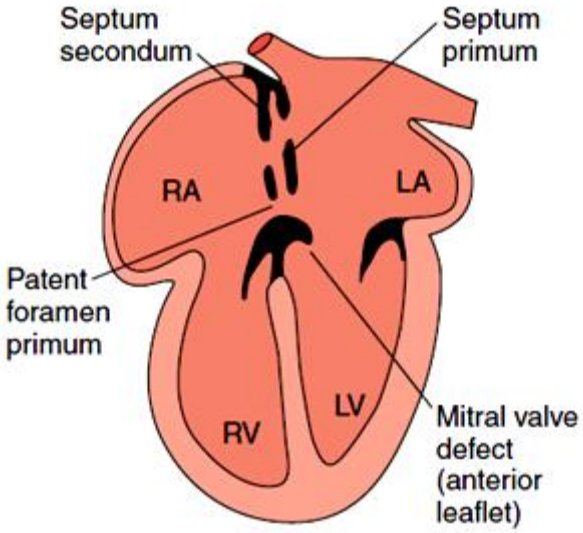
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Anomalies of the atrioventricular canal

1- Persistent A-V canal:

It is usually accompanied by atrial & ventricular septal defect.

Cause: *complete failure of fusion of the ventral and dorsal endocardial cushions.*



2 - Ostium primum defect:

Cause: *partial (incomplete) fusion of the ventral and dorsal endocardial cushions.*

So, septum primum fails to join the septum intermdium -> failure of closure of the ostium primum.

3- Tricuspid atresia:

Cause: *fusion of cusps of the tricuspid valve leading to its absence.*

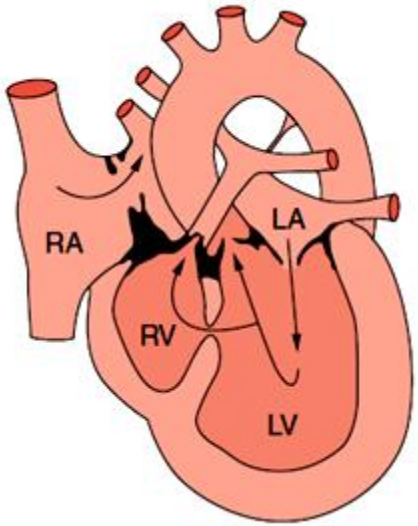
It is usually accompanied by:

1- Patent foramen ovale.

2- Hypertrophy of Lt. ventricle.

3- Patent interventricular foramen

4- Under development of Rt ventricle



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Development of the ventricles

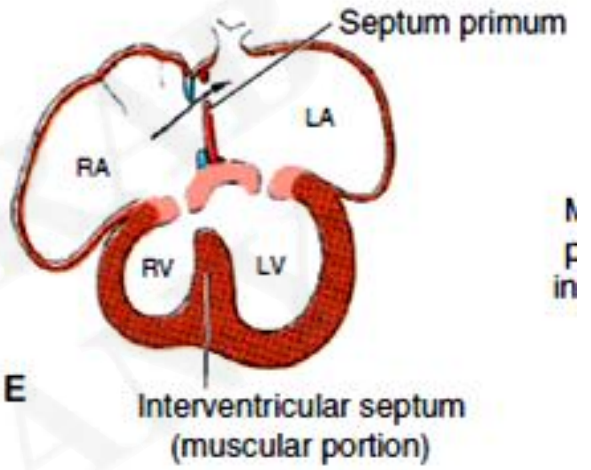
1- Formation of the interventricular septum

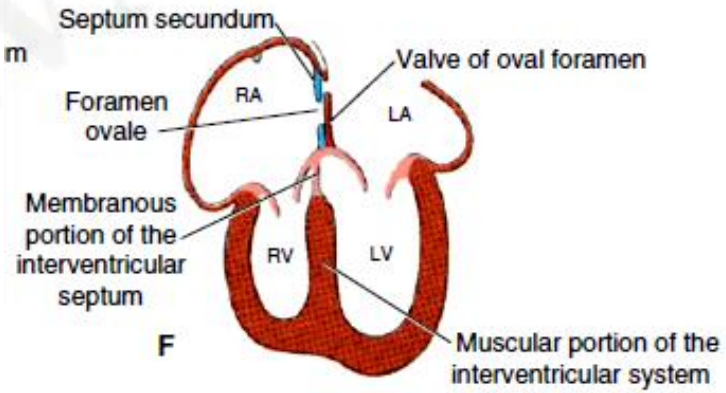
A) Development of muscular part of the interventricular (IV) septum

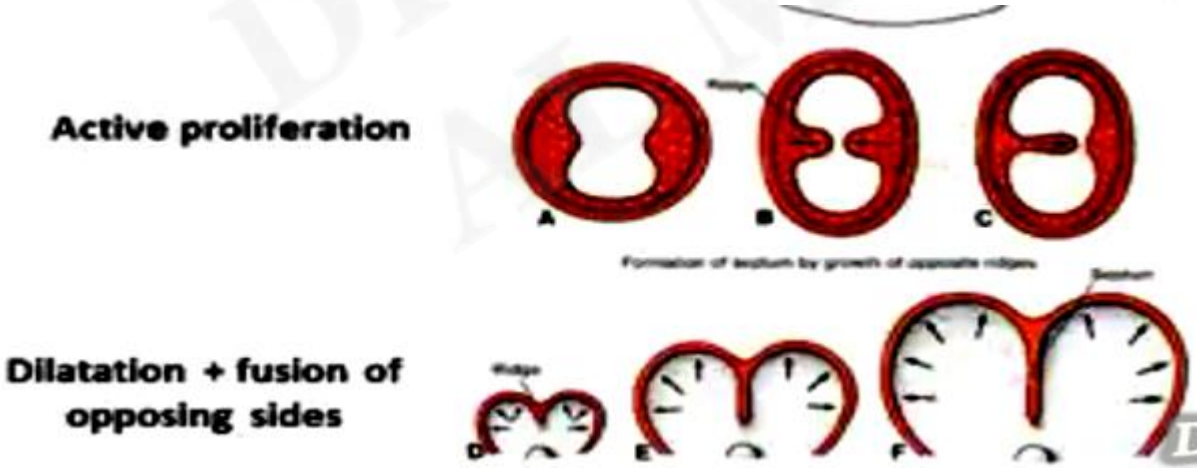
- *The medial walls of the enlarging ventricles approach each other and fuse to form the primordium of the muscular part of the IV septum.*
- *This septum is crescentic in shape and has a concave free edge.*
- *Most of its increase in height results from either:*

1. *Dilation of the ventricles on each side of the septum.*
2. *Active proliferation of myoblasts in the septum.*

- *There is a crescent-shaped interventricular foramen between the free edge of the IV septum & fused endocardial cushions.*







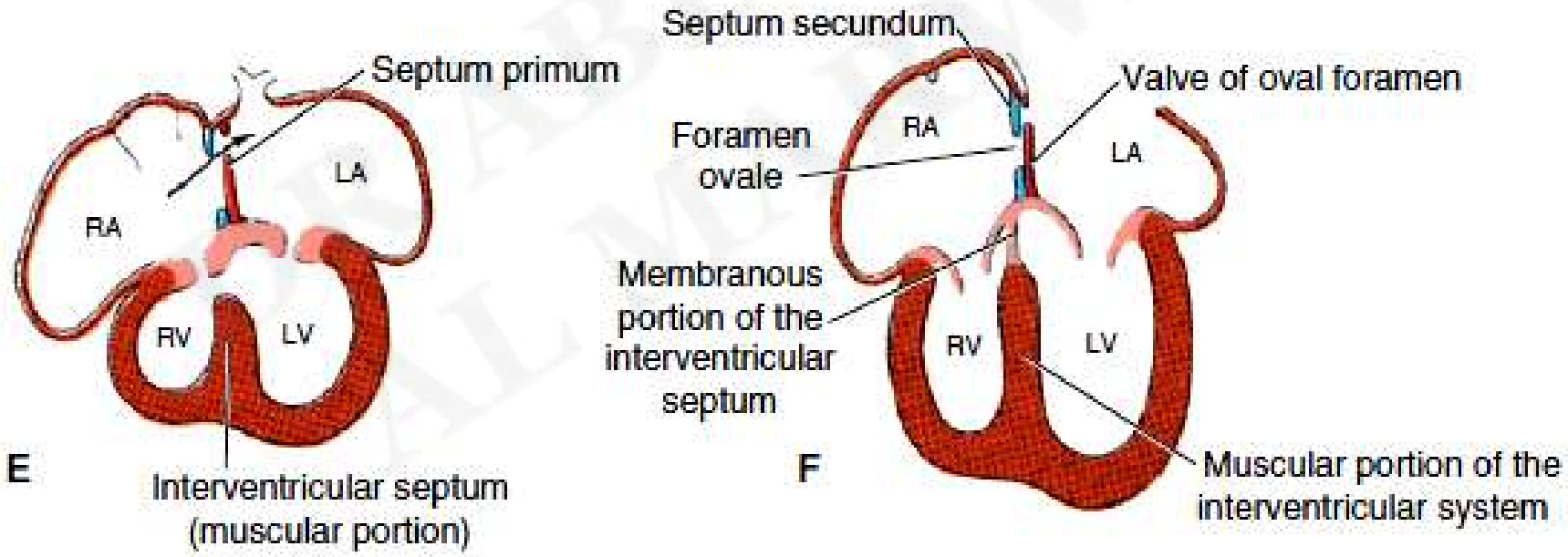
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B) Development of the membranous part of the interventricular septum:

- *This part develops by end of* ***7th week*** *to close the I.V foramen.*

- *It is formed of 2 components:*

1. *Outgrowth of atrioventricular endocardial cushions.*
2. *Lower part of the conus septum (Rt and Lt bulbar ridges)*



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2- Absorption of midportion (conus cordis) of bulbus cordis:

- *Conus cordis is absorbed into the ventricles forming the outflow tracts of both ventricles:*
- *The part absorbed into the **Rt** ventricle will form the **infundibulum**.*
- *The part absorbed into the **Lt** ventricle will form the **vestibule** of the aorta.*



3- Sources of the ventricles

Primitive ventricle: It gives rise to

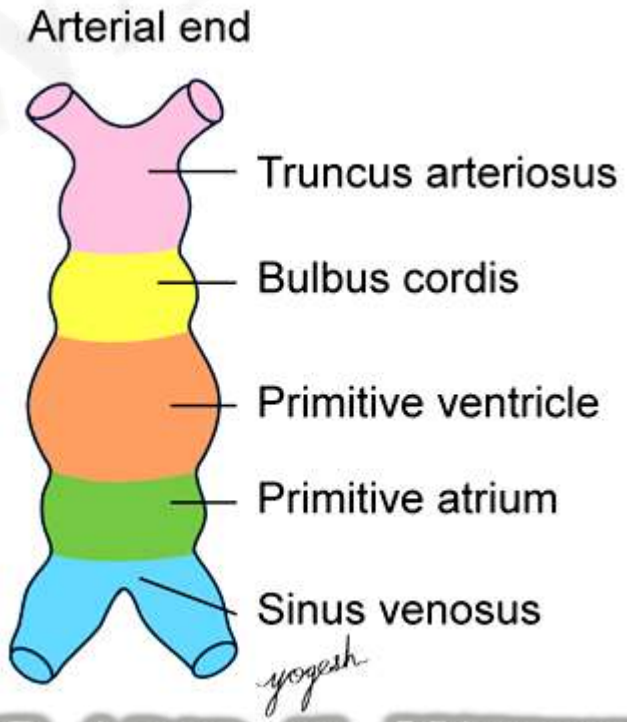
** Most of the left ventricle.*

** Inlet region of the Rt ventricle.*

Bulbus cordis:

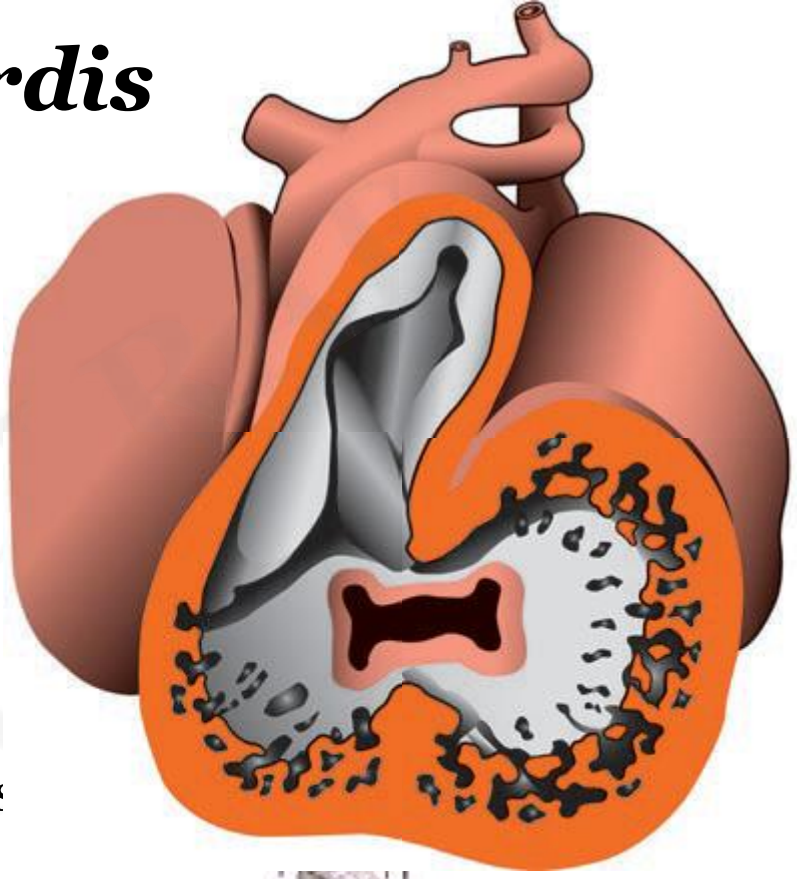
** The proximal part will form most of the right ventricle.*

** The midportion part (conus cordis) forms the outflow tracts of both ventricles.*



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Bulbus Cordis



Parts of the bulbus cordis:

A) **Lower Part :** *Is incorporated with the primitive ventricle (bulboventricular canal).*

This part forms the rough inflowing part of both ventricles.

B) **The Remaining Upper Part :**

i) **The proximal part:** *will form the infundibulum of the* ***right*** *ventricle and the vestibule of the* ***left*** *ventricle.*

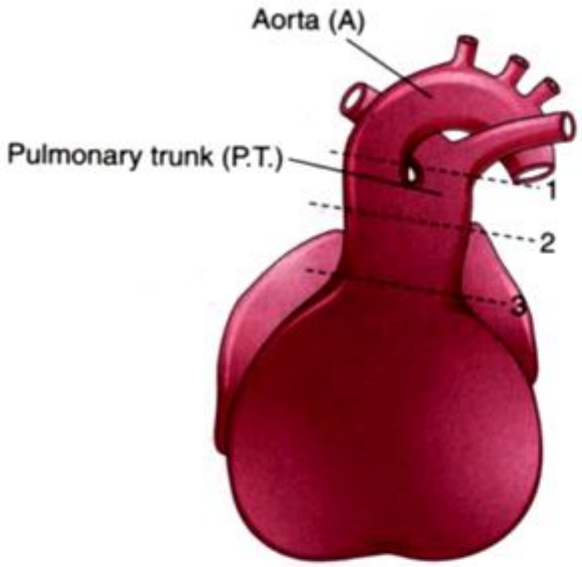
ii) **The mid-portion part (conus cordis):** *is absorbed into the ventricles forming the outflow tracts of both ventricles.*

iii) **The distal part (truncus arteriosus):** *forms the aortic and pulmonary orifices .*

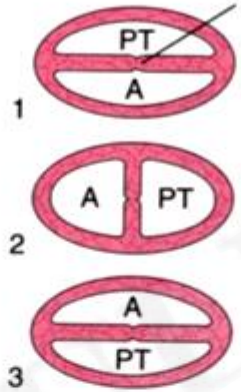
- *It is divided by the spiral aortico-pulmonary septum into roots and proximal portions of the ascending aorta and pulmonary trunk.*



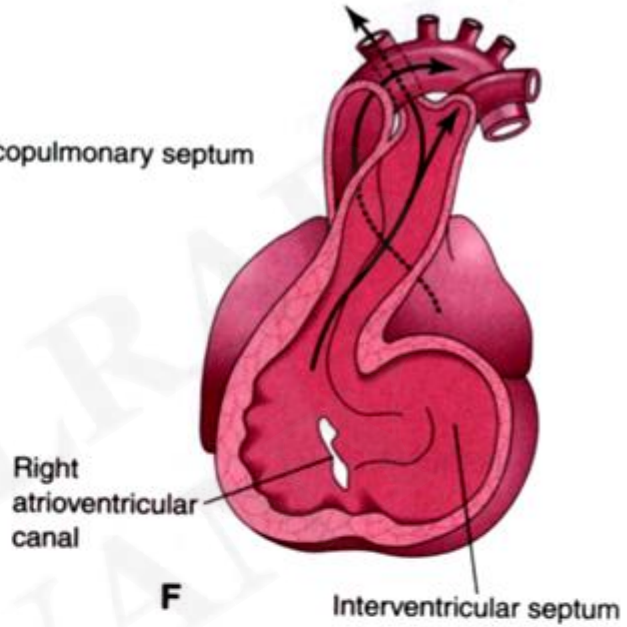
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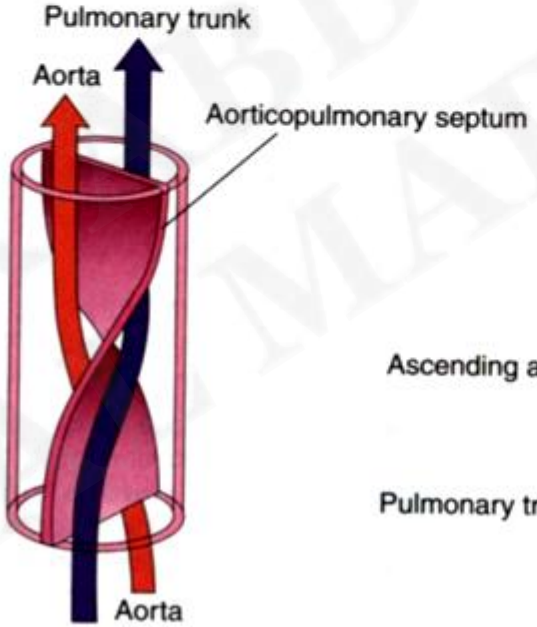
D



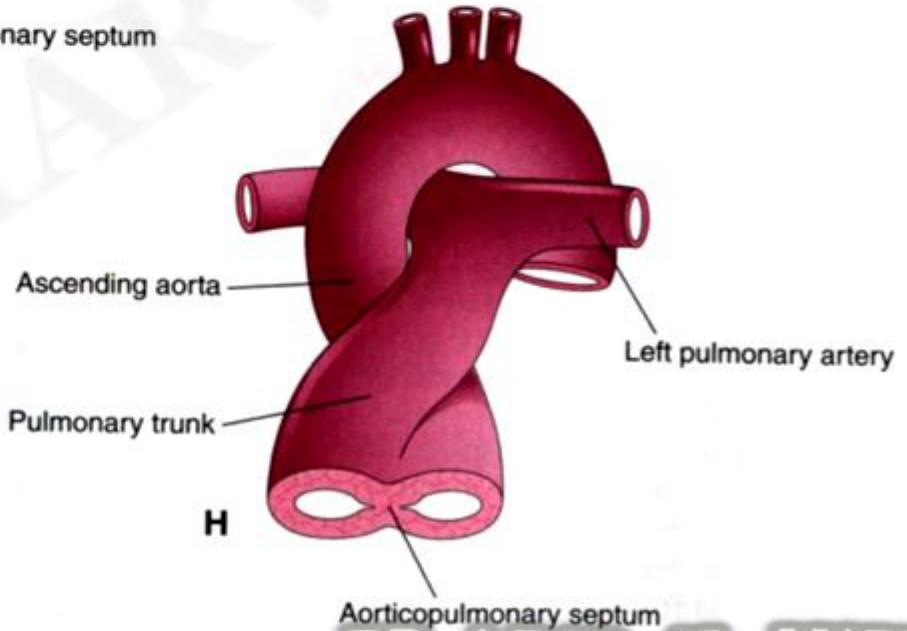
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H

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Truncus arteriosus

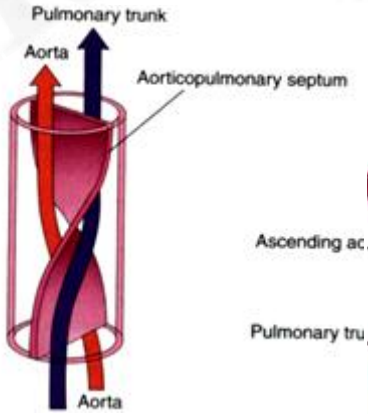
Formation of the aortico-pulmonary septum:

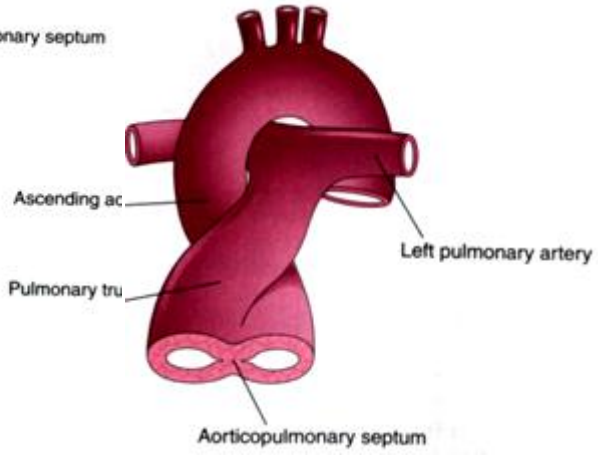
- *During the **5th week**, 2 (Rt & Lt) bulbar and truncal ridges are developed. They are derived from neural crest mesenchyme.*
- *They are **bulbar** (in bulbus cordis) & **truncal** (in truncus arteriosus). They are continuous with each other.*
- *These 2 ridges grow down towards the ventricles in a spiral way so that they become anterior and posterior in middle part of the truncus.*
- *Then, they become Rt and Lt again in the proximal part (near the ventricles) .*
- *By the **8th week**, the 2 ridges become fused to form a spiral aorticopulmonary septum dividing the truncus arteriosus into pulmonary trunk and ascending aorta.*
- *The spiral course of the aortico-pulmonary septum explains:*

  1- Adult relations between the ascending aorta and pulmonary trunk; the pulmonary trunk lies anterior then to the left and finally posterior to the ascending aorta.

  2- The 2 arteries are enclosed in a common tube of visceral pericardium







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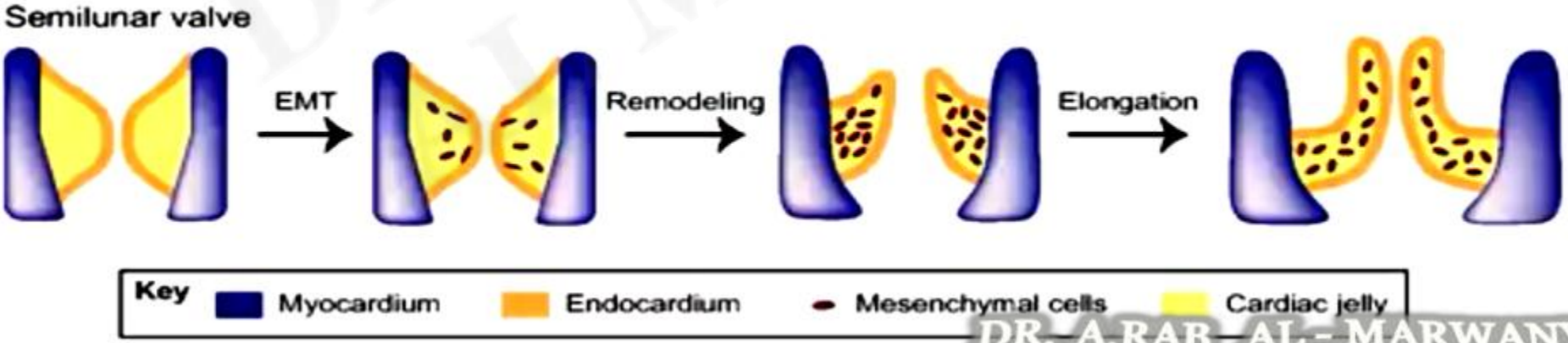
Development of the cardiac valves

- *The semilunar valves develop from three swellings of subendocardial tissue around the orifices of the aorta and pulmonary trunk.*
- *These swellings are hollowed out and reshaped to form three thin-walled cusps.*
- *The atrioventricular (A-V) valves (tricuspid and mitral valves) develop similarly from localized proliferations of tissue around the A-V canals.*

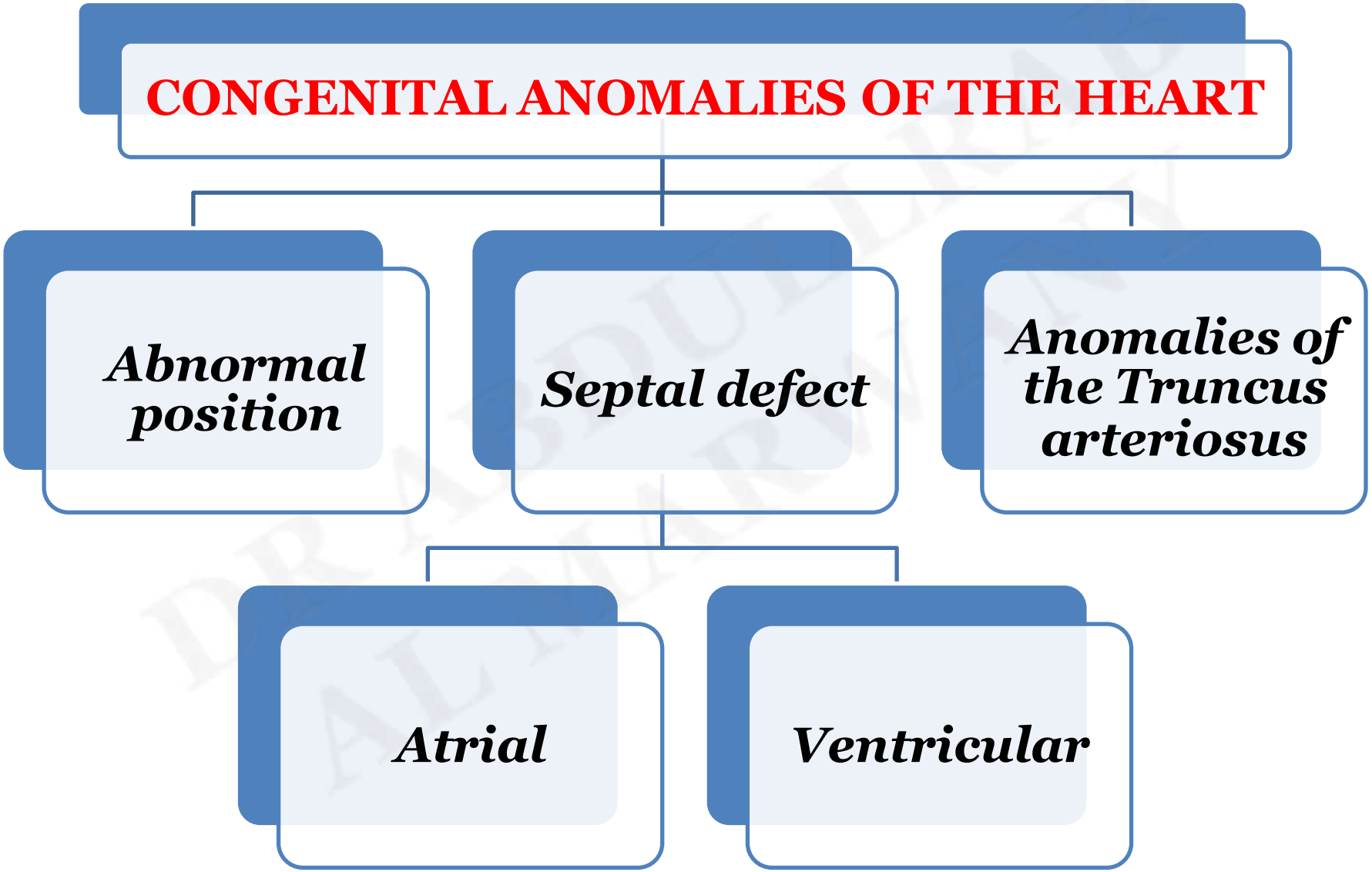
Anomalies of semilunar valves:

1- ***Pulmonary stenosis:*** *is due to fusion the cusp of the pulmonary valve.*

2- ***Aortic stenosis:*** *is due to fusion of the cusps of the aortic valve.*



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CONGENITAL ANOMALIES OF THE HEART
Abnormal position
Septal defect
Anomalies of the Truncus arteriosus
Atrial
Ventricular

Abnormal position

- ***Dextrocardia :*** *right sided heart due to reverse bending of the heart tube.*
- *The heart and its great vessels lie as a mirror image to their normal position.*
- *This may occur alone or may be associated with reversal of all organs (situs inversus totalis).*



- ***Ectopia cordis :*** *protrusion of the heart through a defect in sternum , due to incomplete sternal development.*
- *It is due to failure of the embryo to close in the middle line.*



Septal defect

- ***Anomalies of the interatrial septum***

1·premature Closure Of Foramen Ovale:

Character: *premature closure of foramen ovale in the intrauterine life.*

Results: *hypertrophy of Rt atrium and Rt ventricle with underdevelopment of Lt ventricle. It leads to the death shortly after birth.*

2- Patent Foramen Ovale:

A hole in the middle part of the septum .

Normally, the foramen ovale closes completely at birth.

In 20 - 25 % of people, a probe can be passed through the foramen ovale.

3- Endocardial Cushion ASD :

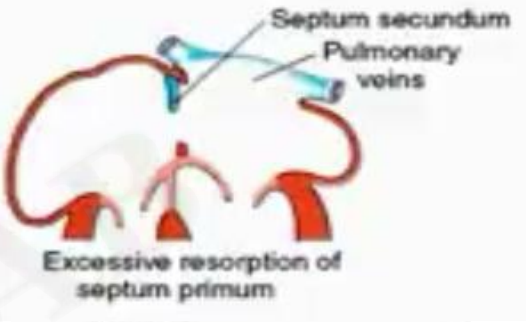
4- Sinus Venosus ASD :

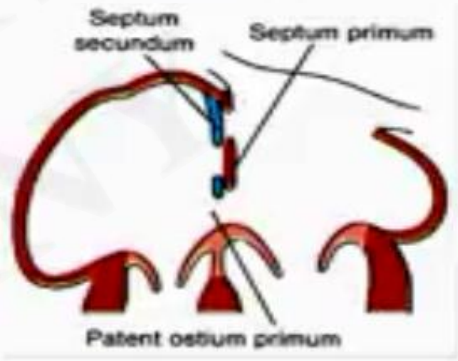
A hole in the lower part of the septum .

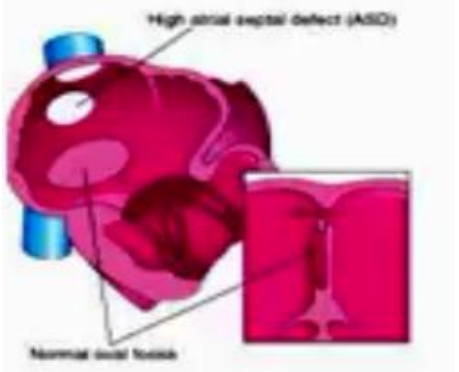
5- Complete failure of formation of the interatrial septum:

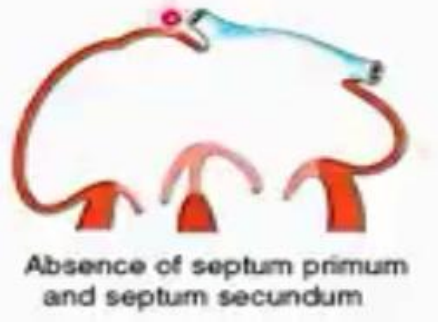
The heart is formed of 3 chambers (common atrium & 2 ventricles) ***Trilocular Heart*** *.*

Cause: *it is due to failure of septation of the primitive atrium*









Anomalies of the interventricular septum (ventricular septal defects).

- ***Membranous Ventricular Septal Defects :*** *A defect in the membranous part of the IV septum.*



- ***Muscular Ventricular Septal Defects :*** *A defect in the membranous as well as the muscular part of the IV septum.*



- ***Common Ventricle :*** ***Trilocular Heart***


Common ventricle

Anomalies of the Truncus arteriosus :



1- Fallot's tetralogy:

Characters:

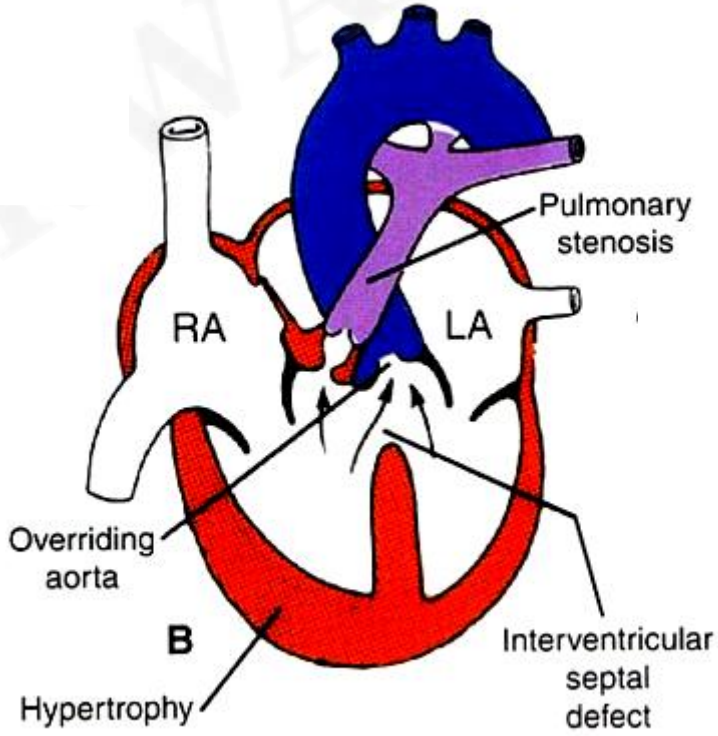
1- Pulmonary stenosis.

2- Hypertrophy of Rt ventricle.

3- Ventricular septal defect.

4- Overriding aorta i.e. the mouth of aorta receives blood from both ventricles.

Cause: anterior displacement of the aortico-pulmonary septum leads to unequal division of the conus.

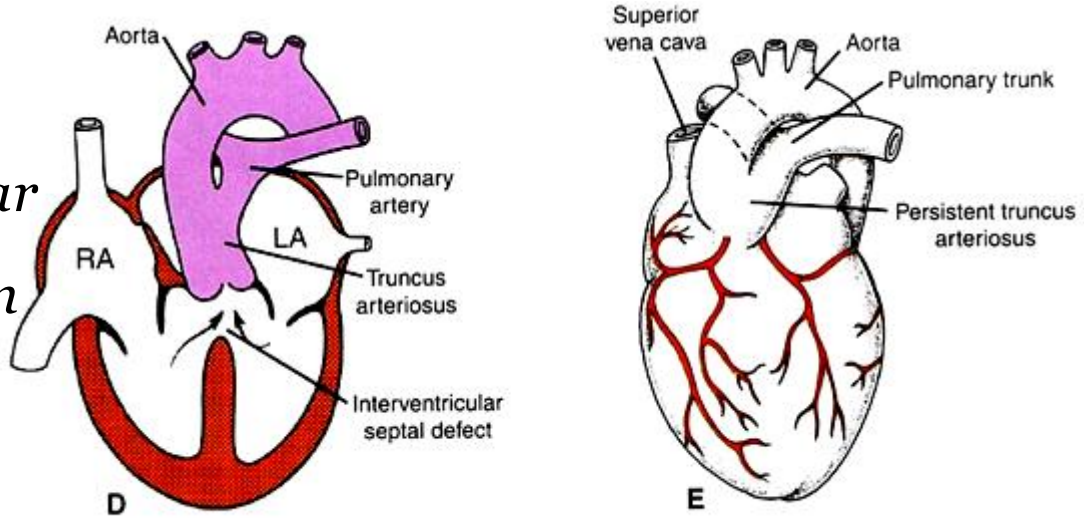


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2- Persistent truncus arteriosus:

Cause: *failure of the 2 bulbar cushions to form the aorticopulmonary septum*

It is accompanied by: *membranous ventricular septal defect.*



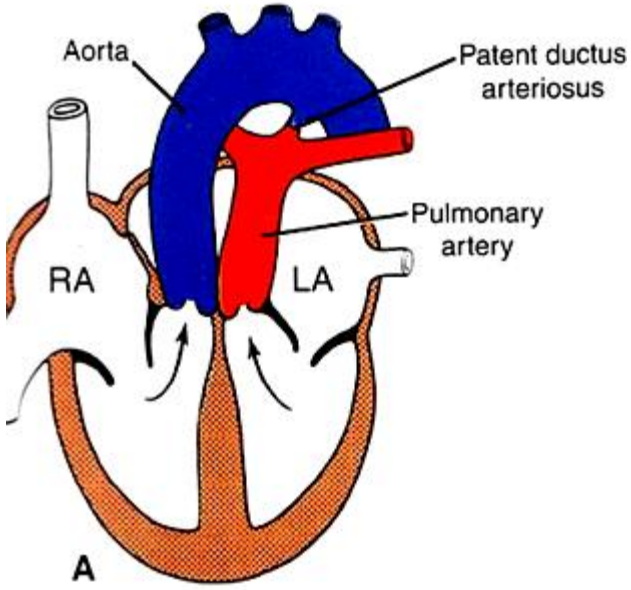
Persistent Truncus Arteriosus

3- Transposition of the Great Arteries (TGA):

Cause: *Aortico-pulmonary septum runs a straight course instead of its normal spiral course.*

It leads to:

* *Aorta is connected below with the Rt. ventricle.*
* *Pulmonary trunk is connected below with the Lt. ventricle*



Transposition of the Great vessels

والى لقاء قادم باذن الله

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